TABLE OF CONTENTS

1 - OVERVIEW AND FLOWCHART

This procedure describes how to calibrate the RFI in a 1.5T system. Please see illustration 1-1 on the next page to confirm that the 1.5T RF/PDU cabinet pictured matches the cabinet to be checked and/or adjusted. The RFI should come from the factory pre-calibrated and, ideally, no initial calibration of the RFI in the field should be needed. The RFI must be checked, however, to confirm that it is calibrated. Calibration is done to prevent faults from improperly balanced RF amplifiers and to make sure that the RFI is outputting the specified 16kW of RF body power and 2kW of RF head power. Adjustments may need to be made if any of the hardware in the RF chain has been replaced (See Appendix B), output levels have decreased with age, or if any of the measurements taken during the check are found not to meet specifications. The head and body RF power levels and balance will be checked in section 2 and, if in specification, no adjustments will be made. The head and body power levels and the phase and gain balance adjustments are made in section 3. Please see the flowchart in Illustration 1-2. Appendices A - G are referred to in the procedure and provide extra information and directions to assist the user with accomplishing the RF power measurement task. Appendix G provides helpful troubleshooting information in the event that a problem is encountered during the RF power check or calibration.

Note

1.5T RF/PDU Replacements or Upgrade Cabinets ONLY: Verify that the External Probe Filter is not connected at MR2A11J1 (located on the inside, rear of the System Cabinet and usually affixed with Velcro to the top of the I/F panel). It was only necessary for CERD exciters with part number 2148300-2. If the filter is present then bypass it. Do this by disconnecting the cable connected on the inside of the cabinet at MR2A1J1 that is coming from the "OUT" port on the filter, disconnecting the cable connected to the "IN" port on the filter, and then reconnecting the cable removed from the filter "IN" port to MR2A1J1 on the inside of the System Cabinet I/F panel. Remove and properly discard the filter and associated cable connected to the filter "OUT" port.

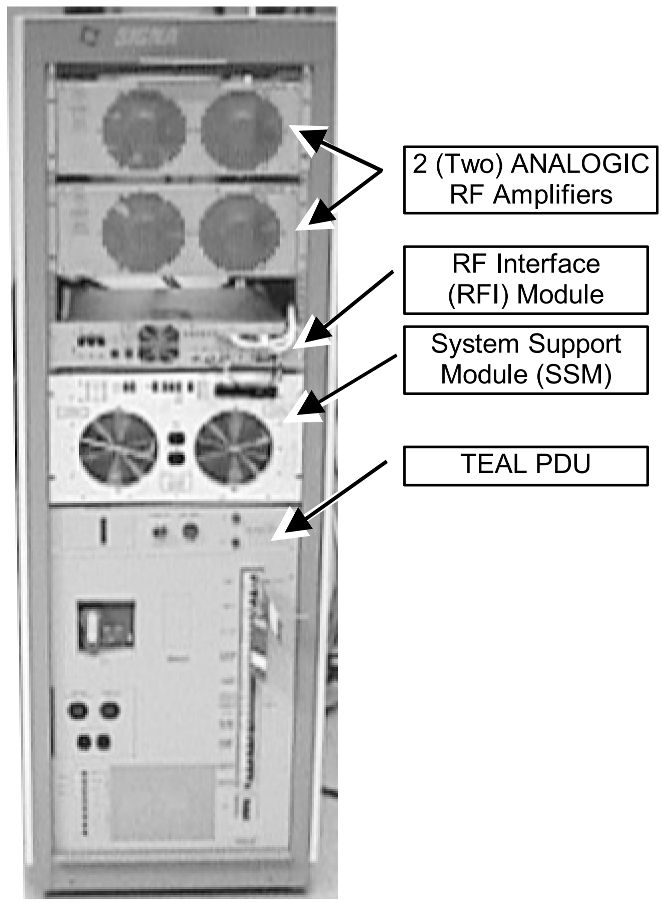
Note

The Amp1 and Amp2 input ports of *early* RF/PDU 1.5T RFI Modules do not have an input impedance of 50 ohms. This is not true of the later modules. *Early* RF/PDU 1.5T RFI Modules can be identified by the vendor manufacturing number of 621000.01 printed in the upper left corner on the rear of the RFI. ONLY units with a manufacturing number of 621000.01 require a special set of impedance matching cables between the rear RFI Amp1 and Amp2 In ports and the respective J4 output ports on the rear of each 1.5T Analogic amplifier. The cables must be connected properly per the attached cable labels and cannot be swapped end to end or the impedance match will be lost. The amplifier end is an RG216 type cable and the RFI end is an RG214 type cable. The RG identifiers are printed on the cable jacket.

1 - OVERVIEW AND FLOWCHART (Continued)

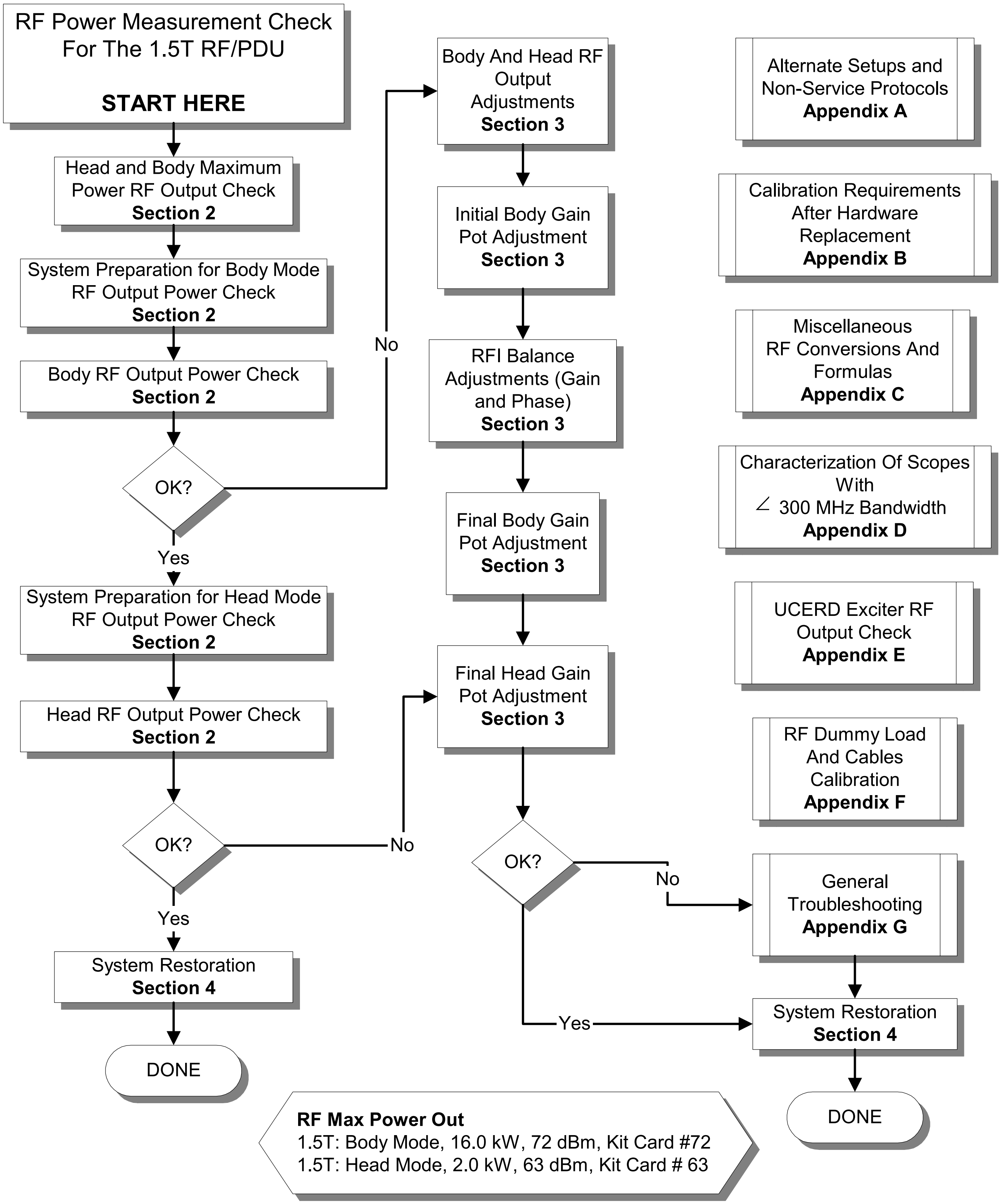
Note

The Body and Head Gain pots are in series with each other. The Body Gain pot is first in the series. The Head Gain pot is factory set so that the head gain is 9 dB less (63 dBm) than the body gain (72 dBm).



1.5T RF/PDU CABINET
ILLUSTRATION 1-1

1 - OVERVIEW AND FLOWCHART (CONTINUED)



CALIBRATION FLOWCHART
ILLUSTRATION 1-2

2 - BODY AND HEAD MAXIMUM POWER RF OUTPUT CHECK

This process will test the assumptions that:

- The (U)CERD Exciter is outputting enough power for the system to meet the rated head and body RF output power levels (Appendix E describes how to check the (U)CERD Exciter output.).
- The RFI has been pre-adjusted at the factory and that, ideally, no further adjustments by the FE are necessary.
- Both RF amplifiers are capable together of outputting 16kW.

The (U)CERD Exciter cannot be adjusted. Appendix E describes how to measure and verify that the Exciter output is correct. The head and body RF output power levels must be checked and, if not correct, adjusted into specification. Three methods are available and listed in the order of preference for measuring head and body RF output power:

- **RF Power Measurement Kit** - Easy to use, <u>preferred</u> method of power measurement.
- **Bird wattmeter** - Must account for cable and load loss to determine actual power.
- **Oscilloscope** - Must know the scope channel correction factor if the scope bandwidth is $<$ 300 MHz. Must account for cable and dummy load loss and the scope bandwidth limitation (channel correction factor) to accurately calculate the power from the observed peak voltage. This method works but the process used to derive the power is somewhat complex and lacks the accuracy of the previous two methods. It is, for this reason, not recommended.

It is *strongly recommended* that the RF Power Measurement Kit be used to obtain an accurate measurement of the RF output power. If it is not possible to obtain an RF Power Measurement Kit then one of the two other methods can be used provided that:

- The Dummy Load and Cables Calibration procedure in Appendix F has been performed and the individual loss values are known.
- The oscilloscope input channel correction factor is known when the scope bandwidth is $<$ 300 MHz *and* the wattmeter is not being used. See Appendix D.

WARNING!

FAILURE TO ACCOUNT FOR THE ACTUAL LOSSES IN THE DUMMY LOAD, CABLES, AND OSCILLOSCOPE WHEN NOT USING THE RF POWER MEASUREMENT KIT TO MEASURE RF POWER CAN RESULT IN SIGNIFICANT MEASUREMENT ERRORS.

A Calibration Flowchart, Illustration 1-2, has been provided in the Overview, Section 1.

All the test points and pots are accessible from the front of the RFI.

2-1 Tools and Instruments Required

Refer to Table 2-1 for equipment required to measure power with the RF Power Measurement Kit or refer to Appendix A (Alternate Equipment Setup). The RF Power Measurement Kit is the preferred method for RF power measurement.

TABLE 2-1
EQUIPMENT REQUIRED IN ADDITION TO THE RF POWER MEASUREMENT KIT

Item	Description	Part Number
1.	100 MHz Scope (equivalent or greater)	46-183029P61
2.	RF Power Measurement Kit	46-317724G1 or G2
3.	50 ohm, 200 Watt, 30dB Attenuator (dummy load)	46-317724P14
4.	Pot Tweaker	46-194427P361
5.	RF Test Cables Kit	46-255816G1
6.	Digital Multimeter (DMM)	46-194427P49
7.	DMM Test Cable – BNC to Banana (optional)	2239139

2-2 System Preparation

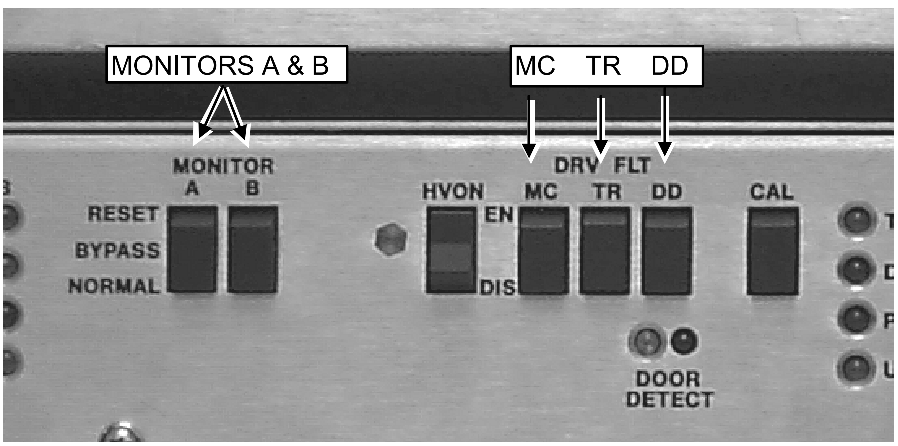
DANGER!!

PROPERTY DAMAGE! PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, BY REMOVING ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

1. Verify that the system is not scanning and that all coils have been removed from the magnet bore. See the **DANGER** message on this page.

2. Remove the front cabinet cover and open the rear cabinet cover.

3. See Illustration 2-1. On the front of the SSM place the:
   - 2 (two) power MONITOR switches (A and B) to the middle BYPASS position.
   - 3 (three) DRV FLT switches to the bottom DIS (disable faults) position.



SSM FRONT PANEL SWITCHES DISABLED
ILLUSTRATION 2-1

4. **If you are using the RF Power Measurement Kit** then calibrate the scope by referring to the RF Power Measurement Kit laminated card set.

   a. Look in the upper right corner of each card and find the card labeled **CAL**.

   b. Configure the scope as in the illustration on the card.

   c. Follow the directions on the card to calibrate the scope using the 4 dBm calibrator.

5. **If you are not using the RF Power Measurement Kit** then complete the Dummy Load and Cables Calibration procedure in Appendix F. If using the scope to measure the RF output power then also make sure that the scope correction factor has been determined for the input channel to be used as described in Appendix D

2-2 System Preparation (Continued)

6. Confirm that the system has not been scanning for at least 10 minutes and then measure the DC voltage present at the BALANCE OUT port J15 on the front of the RFI.

    a. Obtain a DMM and measure the DC voltage (typically about 10 VDC) seen from the BALANCE OUT port J15 on the front of the RFI.

Note
It is convenient to keep the DMM test leads connected to port J15 during the calibration process but it is not necessary. The optional DMM to BNC cable listed in the bottom of Table 2-1 will permit one to easily maintain this connection.

    b. Record the measured voltage in Table 2-2 for later reference.

TABLE 2-2
J15 RFI IDLE BALANCE OUT VOLTAGE

RFI J15 Idle Balance Out Voltage	**________________VDC**

Note
The voltage measured at J15 while the system is idle depends on the ambient room temperature and the individual electrical characteristics of the thermistor in the RFI. It is not necessarily a constant value from one MR system to another.

2-3 Body RF Output Power Check

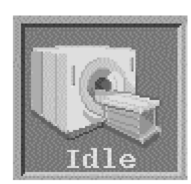
DANGER!!

PROPERTY DAMAGE! PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, BY REMOVING ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

1. Verify that the system is not scanning and that all coils have been removed from the magnet bore. See the two **DANGER** messages on this page.

DANGER!!

PERSONAL INJURY! PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM J3 OR J4 ON THE RFI BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE.



2-3 Body RF Output Power Check (Continued)

2. **If using the RF Power Measurement Kit** then refer to the RF Power Measurement Kit laminated card set.

   a. Look in the upper, right corner of each card and find the card labeled **72** (1.5T Body Output).

CAUTION

The body RF output connection is no longer to the non-existent EFB unit, as the older RF Power Measurement Kit cards show, but instead to the J4 output on the rear of the RFI. An RF adapter is provided in the RF Power Measurement Kit to connect between the HN J4 body RF output and the RF Power Measurement Kit 40 dB N-connector coupler.

   b. Configure the system as shown in the illustration on the card.

   c. Confirm that the rotary attenuator is set to the correct position indicated on the card.

3. **If using the wattmeter or scope** (NOT the RF Power Measurement Kit) to measure power then refer to Appendix A (Alternate Equipment Setup) for the proper system body configuration.

2-3 Body RF Output Power Check (Continued)

4. Prepare the system to scan in Body mode per Table 2-3 or refer to Appendix A (Non-Proprietary protocol).

NOTE
If the system is operating with Release 9.X, CNV4 software, or its equivalent then it is no longer necessary to modify the ia_rf1 and ia_rf2 CV values. These will be set automatically when the calmode CV value is set equal to 5.

TABLE 2-3
SCAN PROTOCOL: BODY MODE

Note: This is the alternate proprietary procedure available for GE use and for sites with a valid Advanced Service Package Limited License.
Refer to Appendix A for the non-proprietary protocol.

A. **[New Pt]**
   Id: **geservice<ENTER>**
   Name: **rfi cals**
   Weight (Lb.): **300<ENTER>**
   Set Patient Protocols to **Service** from the pull-down menu.
B. At front enclosure:
   Landmark in the Head area—remove any coils.
   press **LANDMARK**.
   press **MOVE TO SCAN**.
C. In the Patient Position Protocol field:
   type **o.41.1<ENTER>**(o=Other, 41.1 =series) to load the body protocol
   OR select **other** and select protocol **41** and select series **1**.
D. **[Save Series]**.
E. **[Research Operations]**.
   [Setup Params]. Set TG to **50 [Done]**.
F. **[Research Operations]**.
   [Display CVs]. Highlight CV Name and enter the following:
   CV Name: **calmode<ENTER>**, CV Value: **5<ENTER>** (Dual Logamp Waveform).

NOTE
Skip the next two steps if the system has Release 9.X, CNV4, or equivalent software. The software will set the **ia_rf1** and **ia_rf2** CV values automatically once the calmode CV Value is set equal to 5.

   CV Name: **ia_rf1<ENTER>**, CV Value: **32766<ENTER>** (sets 90º pulse full scale).
   CV Name: **ia_rf2<ENTER>**, CV Value: **0<ENTER>** (turns off 180º pulse).
   [Accept].
G. **[Research Operations]**
H. **[Download]**.

5. Verify Body LED is illuminated on front of RFI Module.

6. **[Manual Prescan] [Scan TR]**. View the Body RF output waveform on the oscilloscope. Slowly increase TG to 200.

2-3 Body RF Output Power Check (Continued)

CAUTION

Check the system center frequency NOW, especially if this is a new installation, and confirm that it consists of 8 digits and that it is correct. The RF amplifiers will output a low and distorted RF waveform if center frequency is wrong due to a missing digit.

7. Verify the Unblank LED's or Gating LED's on the Amplifiers pulse on/off indicating that the amplifiers are outputting RF energy.

8. If the amplifier faults with errors 2225748 or 2225761 (peak power fault) and the cabling to and from the amplifiers is correct, then it will be necessary to calibrate the amplifiers. Proceed to **Section 3 - Body And Head RF Output Power Adjustments**.

9. **If using the RF Power Measurement Kit** then refer to the specified number of divisions needed to achieve 16kW output from the RF Power Measurement Kit card **72** (1.5T Body RF Output) and verify that the waveform displayed on the scope meets, and does not exceed, this specification.

10. **If using the wattmeter procedure** then read the wattmeter display and use the formula below to calculate the body RF power and verify that it meets, and does not exceed, the 16kW (72dBm) specification. The dummy load and cable loss factor was determined from the procedure in Appendix F.

RF Power Measurement (in watts) Using Wattmeter And Formula:
Wattmeter reading (in watts) **X** dummy load and cable loss factor

11. **If using the oscilloscope procedure (NOT the RF Power Measurement Kit)** then read the peak voltage (Vpeak) from the scope display and use the formula below or the Power Calculator spreadsheet (8.X Service Methods CD-ROM must be in the laptop CD-ROM drive) to calculate the RF output power and verify that it meets, and does not exceed, the 16kW (72dBm) specification. The dummy load and cable loss factors were determined from the procedure in Appendix F. The scope correction factor was determined in Appendix D.

RF Power Measurment (in watts) Using Oscilloscope And Formula:
$\dfrac{\left(\dfrac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{100}$ **X** dummy load and cable loss factor

12. Make certain that the RF amplifiers have been pulsing for at least one minute and then use the DMM to measure the DC voltage from the BALANCE OUT J15 port on the front of the RFI. Verify that this voltage doesn't differ from that recorded in Table 2-2 by more than 0.50 VDC.

2-3 Body RF Output Power Check (Continued)

13. Decrease TG to 0 (zero). **[Done] [End Exam]**.

14. Verify that the scan desktop icon is displaying the "Idle" message.

15. If either the body RF power output or the J15 Balance Out voltage did not meet specifications then proceed directly to **Section 3 - Body And Head RF Output Power Adjustments**.

16. If both the body RF power output and the J15 Balance Out voltage met specifications then proceed to the next section, **Section 2-4 - Head RF Output Power Check**.

2-4 Head RF Output Power Check

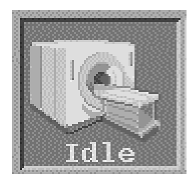
DANGER!!

PROPERTY DAMAGE! PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, BY REMOVING ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

1. Verify that the system is not scanning and that all coils have been removed from the magnet bore. See the two **DANGER** messages on this page.

DANGER!!

PERSONAL INJURY! PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM J3 OR J4 ON THE RFI BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE.



2. **If using the RF Power Measurement Kit** then refer to the RF Power Measurement Kit laminated card set.

    a. Look in the upper, right corner of each card and find the card labeled **63** (1.5T Head Output).

CAUTION

The head RF output connection is no longer to the non-existent EFB unit, as the reference cards in some of the older kits show, but instead to the J3 output on the rear of the RFI.

    b. Configure the system as shown in the illustration on the card.

    c. Confirm that the rotary attenuator is set to the correct position indicated on the card.

2-4 Head RF Output Power Check (Continued)

3. **If using the wattmeter or scope** (NOT the RF Power Measurement Kit) to measure power then refer to Appendix A (Alternate Equipment Setup) for the proper system head configuration.

4. Prepare the system to scan in Head mode per Table 2-4 or refer to Appendix A (Non-Proprietary protocol).

NOTE

If the system is operating with Release 9.X, CNV4 software, or its equivalent then it is no longer necessary to modify the ia_rf1 and ia_rf2 CV values. These will be set automatically when the calmode CV value is set equal to 5.

TABLE 2-4

SCAN PROTOCOL: HEAD MODE

Note: This is the alternate proprietary procedure available for GE use and for sites with a valid Advanced Service Package Limited License.
Refer to Appendix A for the non-proprietary protocol.

A. **[New Series]**
At Patient Protocols – select **other**.
B. In the protocol field, type **o.41.2<ENTER>** (o=Other, 41.2 =series) to load the head protocol **OR** select **[o.41] [Series 2] [Accept]**.
C. **[OK]** (if required).
D. **[Save Series]**
E. **[Prepare to Scan]**
F. **[Research Operations]**.
[Setup Params]. Set TG to **50 [Done]**.
G. **[Research Operations]**.
[Display CVs]. Highlight CV Name and enter the following:
CV name: **calmode<ENTER>**, **5<ENTER>** (Dual Logamp Waveform).

NOTE

Skip the next two steps if the system has Release 9.X, CNV4, or equivalent software. The software will set the **ia_rf1** and **ia_rf2** CV values automatically once the calmode CV Value is set equal to 5.

CV name: **ia_rf1<ENTER>**, **32766<ENTER>** (sets 90º pulse full scale).
CV name: **ia_rf2<ENTER>**, **0<ENTER>** (turns off 180º pulse).
[Accept].
H. **[Research Operations]**
I. **[Download].**

5. Verify Head LED is illuminated on front of RFI Module.

6. **[Manual Prescan] [Scan TR]**. View the Head RF output waveform on the oscilloscope. Increase TG to 200.

2-4 Head RF Output Power Check (Continued)

CAUTION

Check the system center frequency NOW, especially if this is a new installation, and confirm that it consists of 8 digits and that it is correct. The RF amplifiers will output a low and distorted RF waveform if center frequency is wrong due to a missing digit.

7. Verify the Unblank LED's or Gating LED's on the Amplifiers pulse on/off indicating that the amplifiers are outputting RF energy.

8. **If using the RF Power Measurement Kit** then refer to the specified number of divisions needed to achieve 2kW (63dBm) output from the RF Power Measurement Kit card **63** (1.5T Head RF Output) and verify that the waveform displayed on the scope meets, and does not exceed, this specification.

9. **If using the wattmeter procedure** then read the wattmeter display and use the formula below to calculate the head RF power and verify that it meets, and does not exceed, the 2kW (63dBm) specification. The cable loss factor was determined from the procedure in Appendix F.

RF Power Measurement (in watts) Using Wattmeter And Formula:
Wattmeter reading (in watts) **X** cable loss factor

10. **If using the oscilloscope procedure (NOT the RF Power Measurement Kit)** then read the peak voltage (Vpeak) from the scope display and use the formula below or the Power Calculator spreadsheet (8.X Service Methods CD-ROM must be in the laptop CD-ROM drive) to calculate the RF output power and verify that it meets, and does not exceed, the 2kW (63dBm) specification. The dummy load and cable loss factor was determined from the procedure in Appendix F. The scope correction factor was determined in Appendix D.

RF Power Measurement (in watts) Using Oscilloscope And Formula:
$\dfrac{\left(\dfrac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{100}$ **X** dummy load and cable loss factor

11. Decrease TG to 0 (zero). **[Done] [End Exam]**.

12. Verify that the scan desktop icon is displaying the "Idle" message.

13. If the amplifier is not outputting 2kW of RF power but the body power is within specifications then proceed directly to **Section 3-4 - Final Head Gain Pot Adjustment**.

14. If the RF head output power is 2kW and the body output power is 16kW then no further adjustments are necessary. Proceed directly to **Section 4 - Restoration**.

3 - BODY AND HEAD RF OUTPUT POWER ADJUSTMENTS

3-1 Initial Body Gain Pot Adjustment

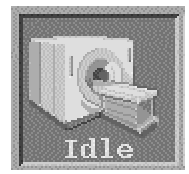
DANGER!!

PROPERTY DAMAGE! PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, BY REMOVING ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

1. Verify that the system is not scanning and that all coils have been removed from the magnet bore. See the two **DANGER** messages on this page.

DANGER!!

PERSONAL INJURY! PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM J3 OR J4 ON THE RFI BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE.



2. **If using the RF Power Measurement Kit** then refer to the RF Power Measurement Kit laminated card set.

   a. Look in the upper, right corner of each card and find the card labeled **72** (1.5T Body Output).

CAUTION

The body RF output connection is no longer to the non-existent EFB unit, as the older RF Power Measurement Kit cards show, but instead to the J4 output on the rear of the RFI. An RF adapter is provided in the RF Power Measurement Kit to connect between the HN J4 body RF output and the RF Power Measurement Kit 40 dB N-connector coupler.

   b. Configure the system as shown in the illustration on the card.

   c. Confirm that the rotary attenuator is set to the correct position indicated on the card.

3. **If using the wattmeter or scope** (NOT the RF Power Measurement Kit) to measure power then refer to Appendix A (Alternate Equipment Setup) for the proper system body configuration.

3-1 Initial Body Gain Pot Adjustment (Continued)

4. Prepare the system to scan in Body mode per Table 3-1 or refer to Appendix A (Non-Proprietary protocol).

NOTE

If the system is operating with Release 9.X, CNV4 software, or its equivalent then it is no longer necessary to modify the ia_rf1 and ia_rf2 CV values. These will be set automatically when the calmode CV value is set equal to 5.

TABLE 3-1
SCAN PROTOCOL: BODY MODE

Note: This is the alternate proprietary procedure available for GE use and for sites with a valid Advanced Service Package Limited License.
Refer to Appendix A for the non-proprietary protocol.

A. **[New Pt]**
Id: **geservice<ENTER>**
Name: **rfi cals**
Weight (Lb.): **300<ENTER>**
Set Patient Protocols to **Service** from the pull-down menu.

B. At front enclosure:
Landmark in the Head area—remove any coils.
press **LANDMARK**.
press **MOVE TO SCAN**.

C. In the Patient Position Protocol field:
type **o.41.1<ENTER>**(o=Other, 41.1 =series) to load the body protocol
OR select **other** and select protocol **41** and select series **1**.

D. **[Save Series]**.

E. **[Research Operations]**.
[Setup Params]. Set TG to **50 [Done]**.

F. **[Research Operations]**.
[Display CVs]. Highlight CV Name and enter the following:
CV Name: **calmode<ENTER>**, CV Value: **5<ENTER>** (Dual Logamp Waveform).

NOTE

Skip the next two steps if the system has Release 9.X, CNV4, or equivalent software. The software will set the **ia_rf1** and **ia_rf2** CV values automatically once the calmode CV Value is set equal to 5.

CV Name: **ia_rf1<ENTER>**, CV Value: **32766<ENTER>** (sets 90º pulse full scale).
CV Name: **ia_rf2<ENTER>**, CV Value: **0<ENTER>** (turns off 180º pulse).
[Accept].

G. **[Research Operations]**

H. **[Download]**.

5. Verify Body LED is illuminated on front of RFI Module.

6. **[Manual Prescan] [Scan TR]**. Slowly increase TG to 200.

3-1 Initial Body Gain Pot Adjustment (Continued)

CAUTION

Check the system center frequency NOW, especially if this is a new installation, and confirm that it consists of 8 digits and that it is correct. The RF amplifiers will output a low and distorted RF waveform if center frequency is wrong due to a missing digit.

Note

If the RF Amplifier trips, adjust the Body Gain pot to reduce the RF output power and then increase the TG. Repeat this process until TG = 200 and the body RF output power is 16kW (72dBm). If the amplifier still trips then set the 30-turn Gain Balance pot to mid-range and the Phase Balance pot fully CCW (Counter-ClockWise). Try adjusting the Body Gain pot and TG again.

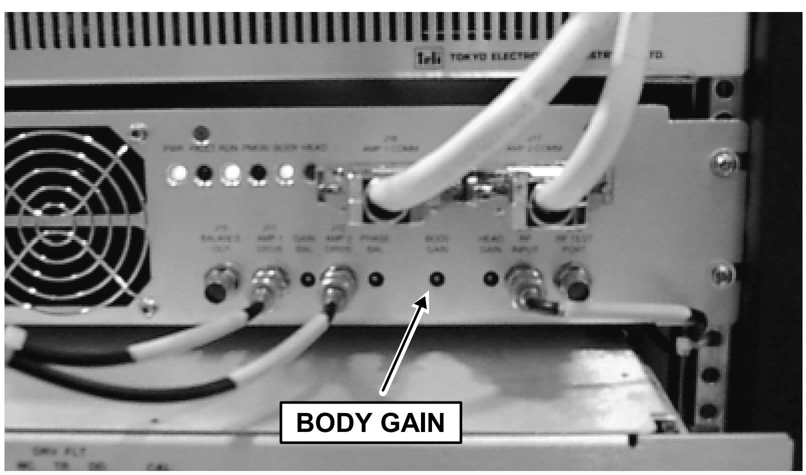
Note

The Power Amplifiers should be capable of 0.3dB to 0.4dB more than the required 72dBm RF Output. Properly balanced systems will trip if the gain is set above 72dBm creating a "Peak Power Fault". This is a FET junction temperature fault which may indicate a poorly balanced RFI or improper gain setting.

Note

The Power Amplifier sampling ports (at the rear) must be 50 ohm terminated to generate full output power and to achieve proper balance.

7. Verify the Unblank LED's or Gating LED's on the Amplifiers pulse on/off.



POT LOCATIONS ON RFI MODULE
ILLUSTRATION 3-1

8. **If using the RF Power Measurement Kit** then refer to the specified number of divisions printed on the RF Power Measurement Kit card **72** (1.5T Body RF Output) needed to achieve 16kW of RF power output. Adjust the Body Gain pot on the front of the RFI so that the RF waveform displayed on the scope meets, but does not exceed, this number of divisions. See Illustration 3-1.

3-1 Initial Body Gain Pot Adjustment (Continued)

9. **If using the wattmeter procedure** then read the wattmeter display and calculate the RF output power from the formula below. The dummy load and cable loss factor was determined from the procedure in Appendix F. Adjust the Body Gain pot on the front of the RFI until the RF output calculated from the formula meets, and does not exceed, the 16kW (72dBm) specification. See Illustration 3-1.

RF Power Measurement (in watts) Using Wattmeter And Formula:
Wattmeter reading (in watts) **X** dummy load and cable loss factor

10. **If using the oscilloscope procedure (NOT the RF Power Measurement Kit)** then read the peak voltage (Vpeak) from the scope display and use the formula below or the Power Calculator spreadsheet (8.X Service Methods CD-ROM must be in the laptop CD-ROM drive) to calculate the RF output power. The dummy load and cable loss factors were determined from the procedure in Appendix F. The scope correction factor was determined in Appendix D. Adjust the Body Gain pot on the front of the RFI until the RF output calculated from the formula meets, and does not exceed, the 16kW (72dBm) specification. See Illustration 3-1.

RF Power Measurement (in watts) Using Oscilloscope And Formula:
$\dfrac{\left(\dfrac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{100}$ **X** dummy load and cable loss factor

11. Decrease TG to 0 (zero). **[Done]**.

12. Proceed to RFI Balance Adjustments (Gain and Phase).

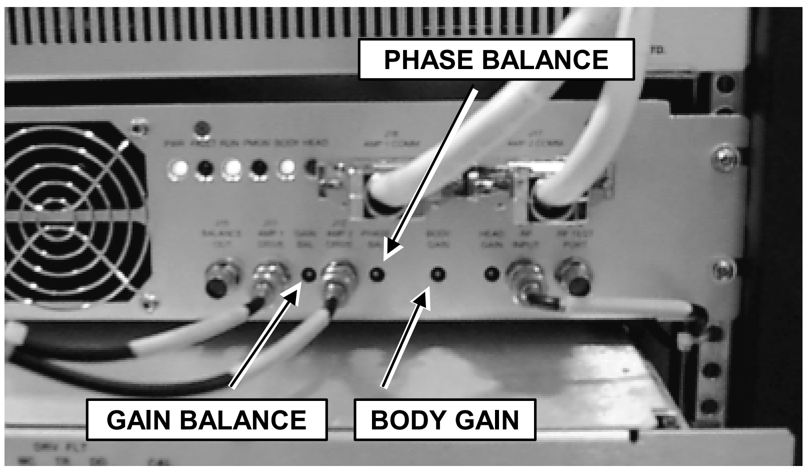
3-2 RFI Balance Adjustments (Gain and Phase)

The Gain Balance and Phase Balance Adjustments are performed using an oscilloscope. These adjustments should be done with the system outputting an amount of RF body output power as close to 16kW (72dBm) as possible.

Note

If the RF Amplifier trips, decrease the RF Output signal by minimizing counter-clock-wise (CCW) the Body Gain Pot Adjustment. If amplitude faults continue to occur, reduce the TG value until the phase and gain adjustments are successfully completed. Check these adjustments again once the RF output power is close to 16kW (72dBm) and the TG is 200.

1. See section 3-1, if necessary, for system setup. Refer to Illustration 3-2 for the pot locations.



BALANCE (PHASE AND GAIN) POT LOCATIONS ON RFI MODULE
ILLUSTRATION 3-2

2. **[Manual Prescan] [Scan TR]**. View the Body RF Output power on the oscilloscope. Slowly increase TG to 200. If the amps keep faulting during this adjustment then the TG can be reduced to a minimum of 170.

3. At the front of the RFI adjust the “Phase Balance” pot fully clock-wise (CW).

4. Adjust the oscilloscope V/div to view the top-most portion of the RF sinc pulse.

5. Adjust the scope to reference the top of the pulse to a horizontal graticle on the display.

Note

Do not confuse the “Body Gain” pot with the “Gain Balance” pot.

6. Set the oscilloscope sweep rate to 500 microseconds / Div. This will expand the RF waveform across the screen and make the following adjustments easier to do.

3-2 RFI Balance Adjustments (Gain and Phase) (Continued)

Note

If the system stops scanning and faults during this adjustment then reduce the TG by five (5) units and restart the manual prescan.

7. At the front of the RFI adjust the “Gain Balance” pot to minimize the RF output sinc pulse and place the center of the pulse it at its lowest value. This is a 30-turn pot and multiple turns of the adjustment screw may be needed before a change is noticed.

8. At the front of the RFI adjust the “Phase Balance” pot counter-clock-wise (CCW) to maximize the RF Output sinc pulse and place the center of the pulse at its highest value. This is a 30-turn pot and multiple turns of the adjustment screw may be needed before a change is noticed.

3-3 Final Body Gain Pot Adjustment And Balance Out Check

The Final Body Gain Pot Adjustment must be performed after the Initial Body Gain, Phase, and Gain Balance adjustments are optimized. The Balance Out voltage at J15 is also checked to make sure that the amplifiers are properly balanced and that neither one is supplying substantially more RF power than the other. The amplifiers should share the workload. A balanced condition between the amplifiers is confirmed by comparing the voltage at J15 while the system is idle (Table 2-2) to that measured after the system has been pulsing for 1 minute. The higher the voltage (the closer it is to the idle voltage), the better balanced the amplifiers. An unbalanced condition will result in a significant decrease (between 0.5 and 1.5 VDC) in the voltage measured at J15 while the system is pulsing. This measurement is affected by ambient room temperature and component tolerances and is not a constant from one RFI to another. This is why two measurements, one taken while the system is idle and one taken while it is operational, are done.

1. **[Manual Prescan] [Scan TR]**. Slowly increase TG to 200.

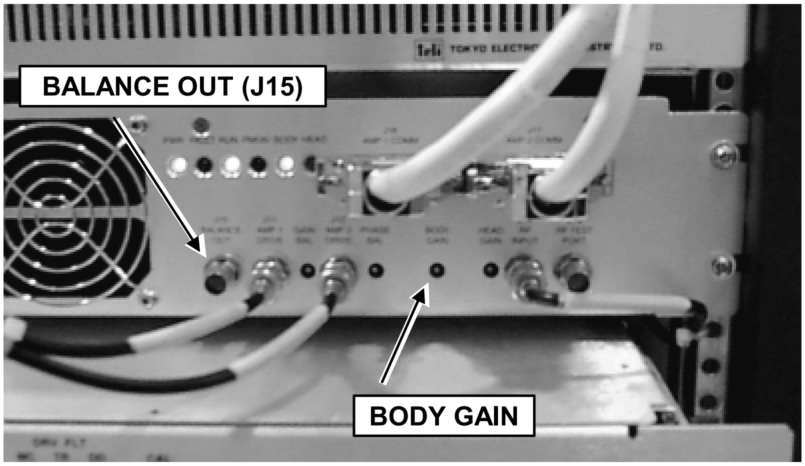
Note

Do not confuse the Body Gain pot with the Gain Balance pot.

Note

If the RF Amplifier trips, adjust the Body Gain pot to reduce the RF output power and then increase the TG. Repeat this process until TG = 200 and the body RF output power is 16kW (72dBm). As long as the calibration in **Section 3-2 RFI Balance Adjustments (Gain and Phase)** was done properly, the Gain and Phase Balance pots should not require any further adjustment.

3-3 Final Body Gain Pot Adjustment And Balance Out Check (Continued)



POT LOCATIONS ON RFI MODULE
ILLUSTRATION 3-3

2. **If using the RF Power Measurement Kit** then refer to the specified number of divisions printed on the RF Power Measurement Kit card **72** (1.5T Body RF Output) needed to achieve 16kW (72dBm) of RF power output. Adjust the Body Gain pot on the front of the RFI so that the RF waveform displayed on the scope meets, but does not exceed, this number of divisions. See Illustration 3-3.

3. **If using the wattmeter procedure** then read the wattmeter display and calculate the RF output power from the formula below. The dummy load and cable loss factor was determined from the procedure in Appendix F. Adjust the Body Gain pot on the front of the RFI until the RF output calculated from the formula meets, and does not exceed, the 16kW (72dBm) specification. See Illustration 3-3.

RF Power Measurement (in watts) Using Wattmeter And Formula:
Wattmeter reading (in watts) **X** dummy load and cable loss factor

4. **If using the oscilloscope procedure (NOT the RF Power Measurement Kit)** then read the peak voltage (Vpeak) from the scope display and use the formula below or the Power Calculator spreadsheet (8.X Service Methods CD-ROM must be in the laptop CD-ROM drive) to calculate the RF output power. The dummy load and cable loss factors were determined from the procedure in Appendix F. The scope correction factor was determined in Appendix D. Adjust the Body Gain pot on the front of the RFI until the RF output calculated from the formula meets, and does not exceed, the 16kW (72dBm) specification. See Illustration 3-3.

RF Power Measurement (in watts) Using Oscilloscope And Formula:
$\dfrac{\left(\dfrac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{100}$ **X** dummy load and cable loss factor

3-3 Final Body Gain Pot Adjustment And Balance Out Check (Continued)

5. Make certain that the RF amplifiers have been pulsing for at least one minute and then use a DMM to measure the DC voltage from the BALANCE OUT J15 port on the front of the RFI. Compare this voltage with that recorded in Table 2-2 and confirm that the difference between the two is not more than 0.50 VDC. Refer to Illustration 3-3 for the location of J15 on the front of the RFI.

6. If the J15 BALANCE OUT voltage fails to meet specification then make sure that the test equipment is not at fault. Repeat the BALANCE OUT and Body Gain Pot Adjustments in this section if necessary. If the J15 BALANCE OUT voltage still fails to meet spec then refer to Appendix G – General Troubleshooting.

7. Decrease TG to 0 (zero). **[Done]**.

8. Verify scan desktop icon is displaying the “Idle” message.

9. Disconnect the DMM from J15 on the RFI.

10. Re-connect Body Heliax after removing test equipment.

11. Proceed to **Section 3-4 – Final Head Gain Pot Adjustment**.

3-4 Final Head Gain Pot Adjustment

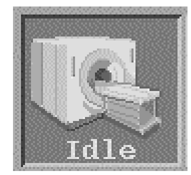
DANGER!!

PROPERTY DAMAGE! PREVENT COIL AND ASSOCIATED SWITCH DAMAGE, BY REMOVING ALL PHANTOMS AND HARDWARE (I.E., HEAD COIL, SURFACE COIL...) FROM THE MAGNET BORE.

1. Verify that the system is not scanning and that all coils have been removed from the magnet bore. See the two **DANGER** messages on this page.

DANGER!!

PERSONAL INJURY! PREVENT POSSIBLE RF BURNS WHEN DISCONNECTING HELIAX CABLES FROM J3 OR J4 ON THE RFI BY VERIFYING THAT THE SYSTEM IS NOT MANUALLY PRESCANNING OR SCANNING. VERIFY THAT THE SCAN DESKTOP ICON DISPLAYS THE "IDLE" MESSAGE.



2. **If using the RF Power Measurement Kit** then refer to the RF Power Measurement Kit laminated card set.

   a. Look in the upper, right corner of each card and find the card labeled **63** (1.5T Head Output).

CAUTION

The head RF output connection is no longer to the non-existent EFB unit, as the reference cards in some of the older kits show, but instead to the J3 output on the rear of the RFI.

   b. Configure the system as shown in the illustration on the card.

   c. Confirm that the rotary attenuator is set to the correct position indicated on the card.

3. **If using the wattmeter or scope** (NOT the RF Power Measurement Kit) to measure power then refer to Appendix A (Alternate Equipment Setup) for the proper system head configuration.

3-4 Final Head Gain Pot Adjustment (Continued)

4. Prepare the system to scan in Head mode per Table 3-2 or refer to Appendix A (Non-Proprietary protocol).

NOTE
If the system is operating with Release 9.X, CNV4 software, or its equivalent then it is no longer necessary to modify the ia_rf1 and ia_rf2 CV values. These will be set automatically when the calmode CV value is set equal to 5.

TABLE 3-2
SCAN PROTOCOL: HEAD MODE

Note: This is the alternate proprietary procedure available for GE use and for sites with a valid Advanced Service Package Limited License.
Refer to Appendix A for the non-proprietary protocol.

A. **[New Series]**
At Patient Protocols – select **other**.
B. In the protocol field, type **o.41.2<ENTER>** (o=Other, 41.2 =series) to load the head protocol **OR** select **[o.41] [Series 2] [Accept]**.
C. **[OK]** (if required).
D. **[Save Series]**
E. **[Prepare to Scan]**
F. **[Research Operations]**.
[Setup Params]. Set TG to **50 [Done]**.
G. **[Research Operations]**.
[Display CVs]. Highlight CV Name and enter the following:
CV name: **calmode<ENTER>**, **5<ENTER>** (Dual Logamp Waveform).

NOTE
Skip the next two steps if the system has Release 9.X, CNV4, or equivalent software. The software will set the **ia_rf1** and **ia_rf2** CV values automatically once the calmode CV Value is set equal to 5.

CV name: **ia_rf1<ENTER>**, **32766<ENTER>** (sets 90º pulse full scale).
CV name: **ia_rf2<ENTER>**, **0<ENTER>** (turns off 180º pulse).
[Accept].
H. **[Research Operations] [Download].**

5. Verify Head LED is illuminated on front of RFI Module.

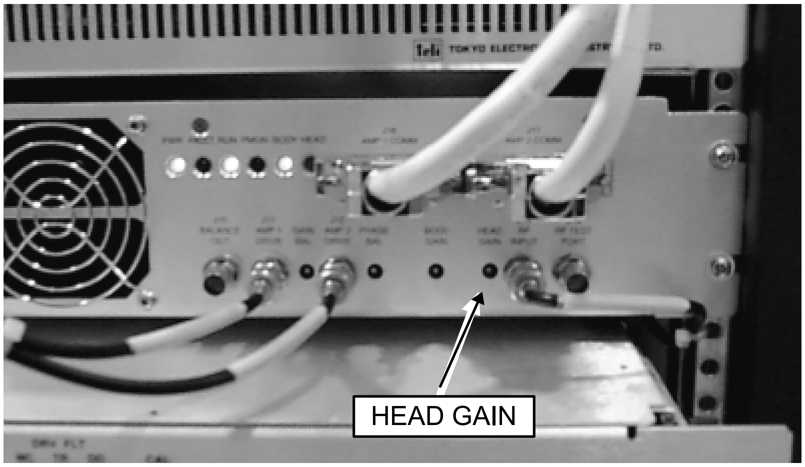
CAUTION

The Head Gain pot is directly affected by changes to the Body Gain pot. Adjust the Head Gain pot ONLY after completing final adjustment of the Body Gain pot.

6. **[Manual Prescan] [Scan TR]**. Increase TG to 200.

7. Verify the Unblank LED's or Gating LED's on the Amplifiers pulse on/off.

3-4 Final Head Gain Pot Adjustment (Continued)



POT LOCATIONS ON RFI MODULE
ILLUSTRATION 3-4

8. **If using the RF Power Measurement Kit** then refer to the specified number of divisions printed on the RF Power Measurement Kit card **63** (1.5T Head RF Output) needed to achieve 2kW of RF power output. Adjust the Head Gain pot on the front of the RFI so that the RF waveform displayed on the scope meets, but does not exceed, this number of divisions. See Illustration 3-4.

9. **If using the wattmeter procedure** then read the wattmeter display and calculate the RF output power from the formula below. The cable loss factor was determined from the procedure in Appendix F. Adjust the Head Gain pot on the front of the RFI until the RF output calculated from the formula meets, and does not exceed, the 2kW (63dBm) specification. See Illustration 3-4.

RF Power Measurement (in watts) Using Wattmeter And Formula:
Wattmeter reading (in watts) **X** cable loss factor

10. **If using the oscilloscope procedure (NOT the RF Power Measurement Kit)** then read the peak voltage (Vpeak) from the scope display and use the formula below or the Power Calculator spreadsheet (8.X Service Methods CD-ROM must be in the laptop CD-ROM drive) to calculate the RF output power. The dummy load and cable loss factor was determined from the procedure in Appendix F. The scope correction factor was determined in Appendix D. Adjust the Head Gain pot on the front of the RFI until the RF output calculated from the formula meets, and does not exceed, the 2kW (63dBm) specification. See Illustration 3-4.

RF Power Measurement (in watts) Using Oscilloscope And Formula:
$\dfrac{\left(\dfrac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{100}$ **X** dummy load and cable loss factor

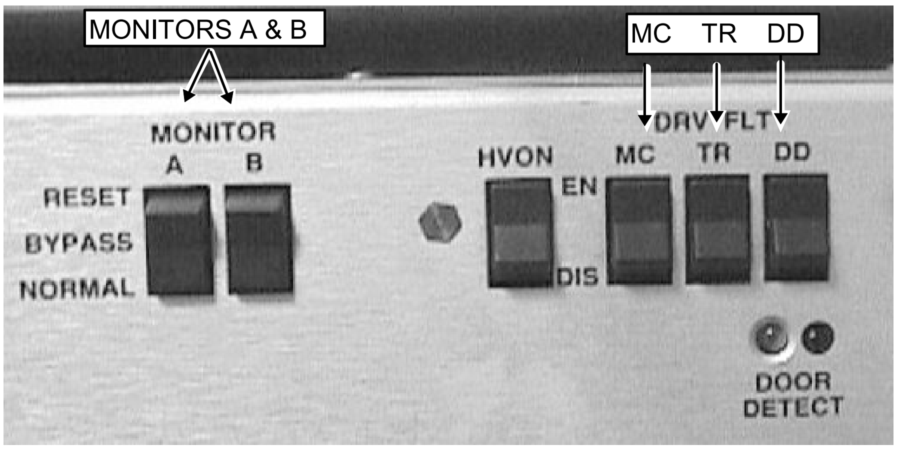
3-4 Final Head Gain Pot Adjustment (Continued)

11. Decrease TG to 0 (zero). **[Done] [End Exam]**.
12. Verify scan desktop icon is displaying the "Idle" message.
13. Replace the RF Head Heliax cable.
14. Proceed to **Section 4 - System Restoration**.

4 - SYSTEM RESTORATION

1. Verify the system is not scanning and the scan desktop icon is displaying the “Idle” message.

2. See Illustration 4-1. On the front panel of the SSM place the:
   - 2 (two) power MONITOR switches (A and B) to the bottom Normal position.
   - 3 (three) DRV FLT switches to the top EN (enable faults) position.



SSM FRONT PANEL SWITCHES ENABLED
ILLUSTRATION 4-1

3. Remove all test equipment.

4. Verify the Body Heliax cable is connected.

5. Verify the Head Heliax cable is connected.

6. Re-install the cabinet covers.

7. Successfully complete one body scan.

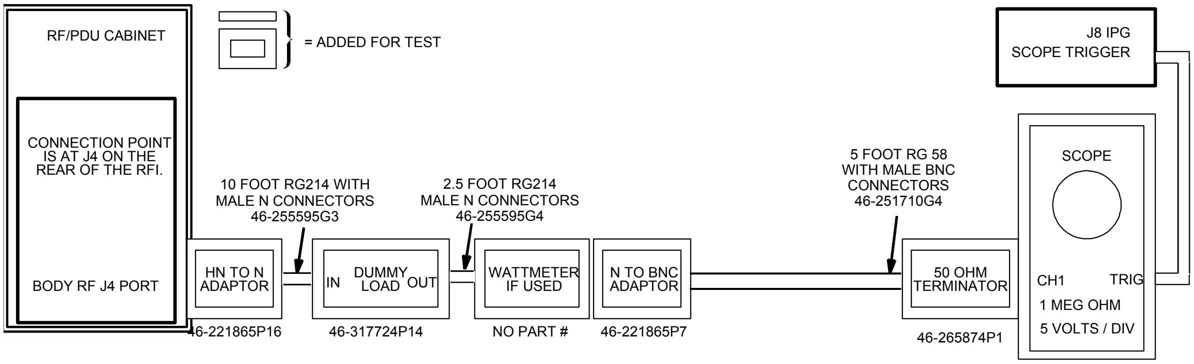
8. Successfully complete one head scan.

APPENDIX A — ALTERNATE EQUIPMENT SETUPS & NON-SERVICE PROTOCOLS

Use this section if the RF Power Measurement Kit is NOT going to be used. This section contains the setup information and protocols necessary to measure the body and head RF output power using either a wattmeter or oscilloscope. If the wattmeter is not used it is strongly recommended that an oscilloscope with a bandwidth of 300 MHz or greater be used for making the RF output power measurements. A 100 MHz oscilloscope can be used to make these measurements **provided** that the oscilloscope input channel to be used has been properly characterized and the resulting correction factor is known. The oscilloscope must be characterized and the correction factor must be known to account for measurement error at 63.86 MHz error due to bandwidth limitations of the 100 MHz oscilloscope.

TABLE A-1
EQUIPMENT REQUIRED IF NOT USING THE RF POWER MEASUREMENT KIT

Item	Description	Part Number
1.	RF Test Cable Kit	46-255816G1
2.	100 MHz Scope (equivalent or greater)	46-183029P61
3.	Digital Multimeter	46-194427P49
4.	50 ohm, 200 Watt, 30 dB Attenuator (dummy load)	46-317724P14
5.	Pot Tweaker	46-194427P361
6.	Wattmeter Kit (optional)	46# not supplied
7.	DMM Test Cable – BNC to Banana (optional)	2239139



ALTERNATE BODY EQUIPMENT SETUP FOR WATTMETER OR SCOPE MEASUREMENT
ILLUSTRATION A-1

Note
If the wattmeter is not available then bypass it's connection in the circuit using an N-female to N-female connector 46-265875P2.

NOTE
If the system is operating with Release 9.X, CNV4 software, or its equivalent then it is no longer necessary to modify the ia_rf1 and ia_rf2 CV values. These will be set automatically when the calmode CV value is set equal to 5.

NON-SERVICE BODY PROTOCOL

PATIENT REGISTER .	**[New Pt]**
PATIENT INFORMATION	
Patient Id	**geservice**
Patient Name	**body rf**
Weight (Lb)	**300** ¬ *IMPORTANT*
	[Landmark]
Landmark	**[>] [Sternal Notch]**
PATIENT PROTOCOLS	**[Patient Position]**
PATIENT POSITION	
Patient Position	**[>] [Supine]**
Patient Entry	**[>] [Head First]**
Coil	**[...] [Body] [Accept]**
IMAGING PARAMETERS	
Plane	**[>] [Axial]**
Mode	**[>] [2D]**
Pulse Seq	**[...] [Spin Echo] [Accept]**
Imaging Options	none (default)
Psd Name	**cal**
Protocol	no entry

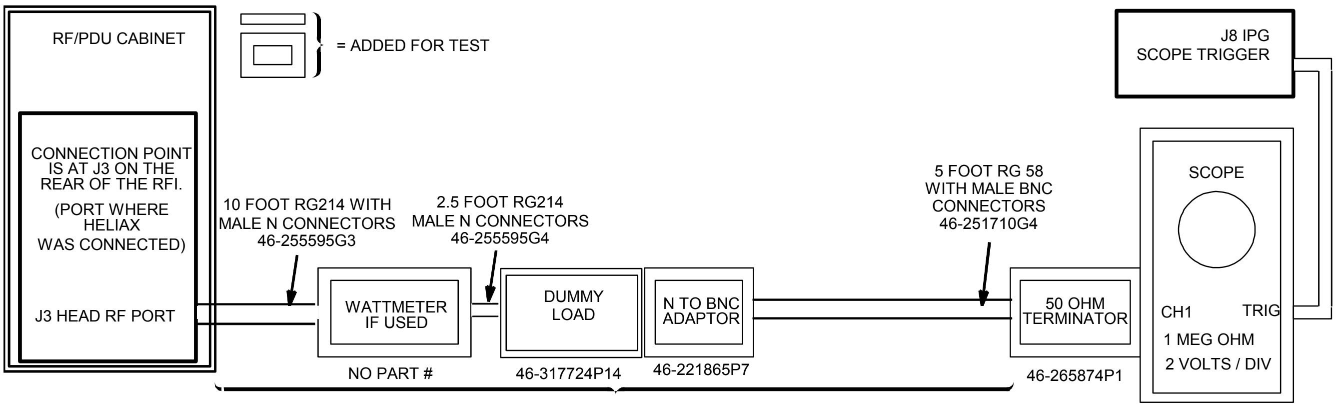
SCAN TIMING	
* of Echoes	1 (default)
TE	**[25]**
TR	**[55]**

SCANNING RANGE	
FOV	**[24]**
Slice Thickness	**[5]**
Spacing	**0**
Start	**0**
End	**0**
# Slices	1 (default)
L/R Center	0 (default)
P/A Center	0 (default)
Table Delta	0.00 (default)
ACQUISITION TIMING	
Freq	**[256]**
Phase	**[128]**
NEX	**[2]**
Freq Dir	**[>] [A/P]**
Auto Center Freq	**[>] [Peak]**
(lowest window)	**[Save Series]**

[Research Operations] [Display CVs]
Modify the following: (if 9.X, CNV4, or equivalent only modify **calmode**)

calmode	**5**
ia_rf1	**32766** (90° maximum)
ia_rf2	**0** (180° minimum)
	[Accept]

[Research Operations] [Setup Params]
Set TG **50 [Done]**
[Research Operations] [Download]
[Prepare to Scan]



ALTERNATE HEAD EQUIPMENT SETUP
ILLUSTRATION A-2

Note
If the wattmeter is not available then bypass it's connection in the circuit using an N-female to N-female connector 46-265875P2.

NOTE
If the system is operating with Release 9.X, CNV4 software, or its equivalent then it is no longer necessary to modify the ia_rf1 and ia_rf2 CV values. These will be set automatically when the calmode CV value is set equal to 5.

NON- SERVICE HEAD PROTOCOL

PATIENT REGISTER .	**[New Pt]**
PATIENT INFORMATION	
Patient Id	**geservice**
Patient Name	**head rf**
Weight (Lb)	**300** ¬ *IMPORTANT*
	[Landmark]
Landmark	**[>] [Sternal Notch]**
PATIENT PROTOCOLS	**[Patient Position]**
PATIENT POSITION	
Patient Position	**[>] [Supine]**
Patient Entry	**[>] [Head First]**
Coil	**[...] [Head] [Accept]**
IMAGING PARAMETERS	
Plane	**[>] [Axial]**
Mode	**[>] [2D]**
Pulse Seq	**[...] [Spin Echo] [Accept]**
Imaging Options	none (default)
Psd Name	**cal**
Protocol	no entry
SCAN TIMING	
* of Echoes	1 (default)
TE	**[25]**
TR	**[55]**

SCANNING RANGE	
FOV	**[24]**
Slice Thickness	**[5]**
Spacing	**0**
Start	**0**
End	**0**
# Slices	1 (default)
L/R Center	0 (default)
P/A Center	0 (default)
Table Delta	0.00 (default)
ACQUISITION TIMING	
Freq	**[256]**
Phase	**[128]**
NEX	**[2]**
Freq Dir	**[>] [A/P]**
Auto Center Freq	**[>] [Peak]**
(lowest window)	**[Save Series]**

[Research Operations] [Display CVs]
Modify the following: (if 9.X, CNV4, or equivalent only modify **calmode**)

calmode	**5**
ia_rf1	**32766** (90° maximum)
ia_rf2	**0** (180° minimum)
	[Accept]

[Research Operations]	**[Setup Params]**
Set TG	**50 [Done]**
[Research Operations]	**[Download]**
[Prepare to Scan]	

APPENDIX B — CALIBRATION REQUIREMENTS AFTER HARDWARE REPLACEMENT

TABLE B-1
RE-CALIBRATION REQUIREMENTS AFTER HARDWARE REPLACEMENT

Hardware Replaced	Required Calibration
1. Exciter Board ((U)CERD)	1. Entire procedure
2. RF Power Amplifier Module	2. Balance and Gain Adj.
3. RFI Module	3. Entire Procedure
4. BNC cable that drives the RF Power Amps.	4. Balance and Gain Adj.
5. N cable between RF Power Amps. And RFI	5. Balance and Gain Adj.

APPENDIX C — SOFTWARE AND FORMULAE FOR MISCELLANIOUS RF CONVERSIONS

Software are described and formulae are presented in this section that will allow the FE to perform various conversions and calculations.

C-1 RF Conversion Software

C-1-1 RF Power Calculator

The first tool, the RF Power Calculator, is a UNIX-based application available on the LX system under the **Toolbelt icon**, **[Utilities]**, **[RF Calculator], [Start]**. Watts to dBm, dBm to Watts, relative volts to relative power, and relative power to relative volts conversions can be done with the RF Power Calculator tool.

C-1-2 Power Calculator Spreadsheet

The second tool, the Power Calculator, is an MS Excel spreadsheet that is available from the 8.X Service Methods CD-ROM. The 8.X Service Methods CD-ROM must be in the laptop CD-ROM drive to start this tool. Manually start the tool from a laptop equipped with MS Excel and running MS Windows 95 or 98 by selecting **Start, Run,** and then entering the following file pathname in the "Open" entry box (include the quotation marks with the pathname): "E:\troubsht\rf\testflow\power calc.xls" . Next, select **OK**. The tool is divided into 3 sections. The first section allows one to calculate RF power from measured peak voltage when using a 100 MHz oscilloscope and a properly characterized RF cables kit and dummy load. The second section provides Vpp to dBm, dBm to Watts, and dBm to Vpp conversion capability. The third section provides short instructions and a means for comparing the attenuation in dB reported on each RF Power Measurement Kit Card with the measured Magnitude Squared Attenuation Factor reported by the Attenuation Test tool that is described in **APPENDIX D – DUMMY LOAD AND CABLES CALIBRATION.**

C-2 RF Conversion Formulae

These are provided to assist the experienced FE with RF power measurement and troubleshooting. The RF conversions below are presented in two different formats. The first format is the actual mathematical formula. The second format assumes that you are entering the data into the LX system calculator exactly as you read it (Access the LX system calculator by right mouse clicking in the background and then selecting the calculator from the Root menu.).

<u>VP-P to dBm Calculation:</u>

$$\text{dBm} = 20 \log \left(\frac{Vpp}{0.632} \right)$$

VP-P **[÷]** 0.632 **[=]** **[LOG]** **[*]** 20 **[=]** dBm.

APPENDIX C — SOFTWARE AND FORMULAE FOR MISCELLANIOUS RF CONVERSIONS (CONTINUED)

dBm to VP-P Calculation:

Vpp = $10^{\left(\frac{dBm}{20}\right)}$ **X** 0.632 or Vpp = antilog $\left(\frac{dBm}{20}\right)$ **X** 0.632

dBm **[÷]** 20 **[=]** **[INV LOG]** **[∗]** 0.632 **[=]** VP-P.

Example: Using 3.65 dBm, the Scope Power reading listed on a particular Kit card, and the LX system calculator to determine the equivalent Vpp:

3.65 **[÷]** 20 **[=]** (0.1825) **[INV LOG]** (1.5222991) **[∗]** 0.632 **[=]** (0.9620931) VP-P.

dBm to Watts Calculation:

Watts = $10^{\left(\frac{dBm}{10}\right)}$ **X** 0.001 or Watts = antilog $\left(\frac{dBm}{10}\right)$ **X** 0.001

dBm **[÷]** 10 **[=]** **[INV LOG]** **[=]** **[∗]** 0.001 **[=]** Total Watts.

C-3 1.5T Division Conversions

The 1.5T division conversion below may be useful for the FE using the RF Power Measurement Kit and attempting to measure a signal that requires more resolution that the "division" unit provides. These examples are provided as a general reference and are not intended to be used to perform the power monitor check.

1.5T Division Conversion example:

- 1.5T sites using the 1.5T Scope Calibrator and 100 MHz Oscilloscope equipment:

$$\frac{\text{1.00 VPP (4.0 dBm)}}{\text{Set to 8 Major Div}} = \frac{\text{0.125 VPP each Major Division}}{\text{5 Minor Divisions}} = \text{0.025 VPP each Minor Division}$$

$$\text{0 dBm} = \frac{\text{0.632 VPP}}{\text{0.025 VPP each Minor Division}} = \textbf{25.28 Minor Divisions}$$

25.28 Minor Divisions = 5 Major Divisions and .28 of a Minor Division

APPENDIX D — CHARACTERIZATION FOR SCOPES WITH < 300 MHZ BANDWIDTH

WARNING!

THIS PROCEDURE REQUIRES THAT THE SCOPE IN USE HAS A BANDWIDTH OF AT LEAST 100 MHZ. DO NOT ATTEMPT TO PERFORM RF POWER MEASUREMENTS WITH AN OSCILLOSCOPE THAT HAS A BANDWIDTH LESS THAN 100 MHZ. SIGNIFICANT MEASUREMENT ERRORS CAN RESULT.

The scope correction factor for the scope channel in use must be known for oscilloscopes with a bandwidth < 300 MHz. One may assume that the correction factor is 1 (unity) for oscilloscopes with a bandwidth ≥ 300 MHz. Do not assume a correction factor for an input channel and do not assume that all the scope input channels have the same correction factor. Each channel must be characterized separately. There are three ways that this can be done.

D-1 Scope Channel Characterization Done During Calibration

One way that the scope channel characterization can be accomplished is to request this information from the metrology lab the next time the scope is sent back for calibration. Attach a tag in a conspicuous location on the scope before it is sent back for calibration requesting this information. Ask that the percent of signal loss at 63.86 MHz (1.5T) and 42 MHz (1.0T) be measured at the 5 volt/div and 2 volt/div settings (for body and head measurements, respectively) for both scope channels and that this information be included with the returned scope. The metrology lab can usually do this upon request and without charge during the calibration process.

D-2 Scope Channel Characterization Done By Comparison

A 400 MHz scope exhibits negligible measurement error at 63.86 MHz or 42 MHz. If a 400 MHz scope is available then it can be compared side-by-side on site with a scope of < 300 MHz bandwidth. Be sure the scope inputs have a 1Mohm impedance and then are terminated with a 50 ohm feed-through adapter before making the measurements. Be sure to characterize all the inputs that are to be used. The signal loss at 63.86 MHz (1.5T) and 42 MHz (1.0T) for each channel at the 5 volt/div and 2 volt/div settings can be determined from the ratio of signal measured on the 100 MHz scope to that seen on the 400 MHz scope

$$\frac{\text{100 MHz scope measured Vpp signal}}{\text{400 MHz scope measured Vpp signal}} = \text{scope correction factor}$$

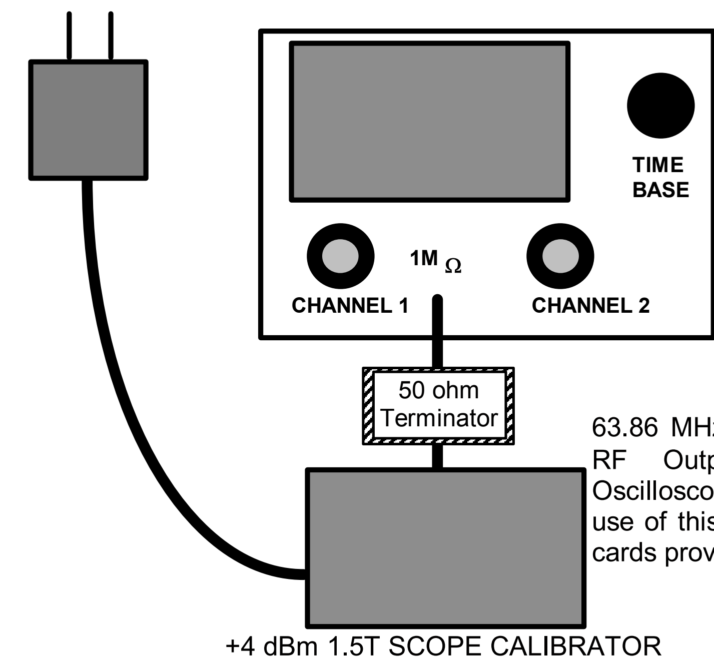
. As long as the same cables are used the loss can also be determined for each cable.

D-3 Scope Characterization Using the RF Power Measurement Kit Signal Reference

The RF Power Measurement Kit contains a 63.86 MHz, 4dBm (1 Vpp) calibrated signal reference that can be used to characterize the scope input channels. Be aware that these values are accurate only for 1.5T systems.

D-3 Scope Characterization Using the RF Power Measurement Kit Signal Reference (Continued)

1. Connect the calibrated reference to the scope as in Illustration D3-1 below.



Note
63.86 MHz oscillator with 4dBm/1.00 VP-P RF Output into 50 Ω termination. Oscilloscopes less than 300 MHz require the use of this 1.5T Scope Calibrator. All 1.5T cards provide Scope Readings in Divisions.

1.5T SCOPE CALIBRATOR CONNECTIONS
ILLUSTRATION D3-1

2. Set the scope amplitude for the channel to be checked to 0.2 volts/div.
3. Set the sweep rate to 5 nsec.
4. Confirm that the channel input impedance is set for 1Mohm and not 50 ohms.
5. Confirm that the scope bandwidth limit switch is not enabled.
6. Confirm that the "Uncal" knob is not rotated and the channel is not uncalibrated.
7. Carefully measure the peak to peak voltage from the calibrated source. The measured value should be 1 Vpp or less.
8. Calculate the correction factor (it should be less than 1):

$$\frac{\text{Vpp measured}}{1\,\text{Vpp}} = \text{correction factor}$$

9. Repeat this process and derive the correction factor for the other scope input channel(s).

D-3 Scope Characterization Using the RF Power Measurement Kit Signal Reference (Continued)

10. Note these correction factors for each channel on a piece of tape affixed to the scope for later reference.

APPENDIX E — (U)CERD EXCITER RF SIGNAL OUTPUT

A minimum RF signal level from the (U)CERD Exciter is required or else it will be impossible to meet the specified 16kW body and 2kW head RF power output levels. Use this section to verify that the (U)CERD Exciter is outputting at least the minimum amount of power required to the RFI. See Table E-1 for needed tools and test equipment.

TABLE E-1
ITEMS NEEDED TO CHECK (U)CERD RF SIGNAL OUTPUT

Item	Description	Part Number
1.	RF Test Cable Kit	46-255816G1
2.	Cannon to BNC-male Test Cable (included in item 4 below)	46-301549P6
3.	SMB-female to BNC-male Test Cable (included in item 4 below)	46-301549P5
4.	TPS RF Connector/Adapter and Cable Test Kit (optional)	46-301927G1
5.	RF Power Measurement Kit (recommended)	46-317724G1 or G2
6.	100 MHz Scope (equivalent or greater)	46-183029P61

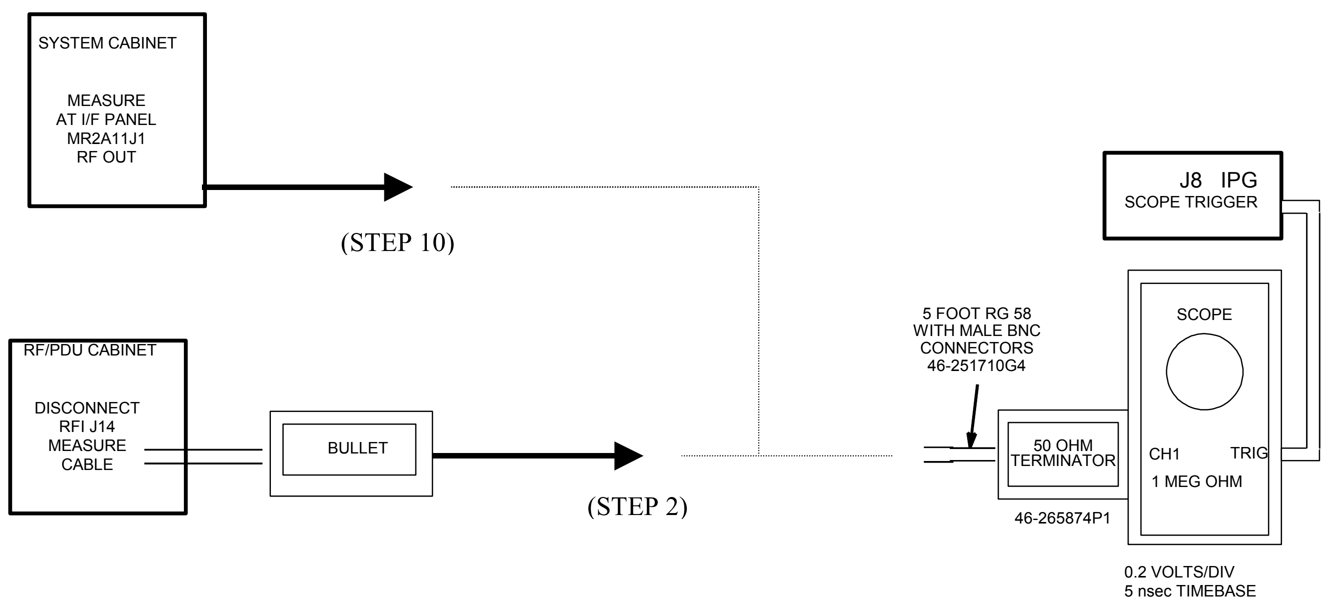
Note

Please read the following if you do not have an RF Power Measurement Kit and you are going to attempt to measure the exciter output directly with a 100 MHz oscilloscope. Be aware that, due to oscilloscope bandwidth limitations, an exciter RF voltage reading taken directly from a 50 ohm terminated 100 MHz oscilloscope input channel can be as much as 16% less than the actual exciter RF output voltage level. This error is accounted for and is not a problem when using the oscilloscope in conjunction with the RF Power Measurement Kit. Reading *reasonably* accurate 63.86 MHz voltage levels directly from a 100 MHz oscilloscope *without* using the RF Power Measurement Kit requires that the correction factor of the scope channel being used is known. This is so that the approximate true RF voltage level can be calculated. The FE can request that the 63.86 MHz correction factor for each channel be determined and reported by the metrology lab the next time the oscilloscope is returned for calibration or, if the FE has access to the 4dBm (1 Vpp) oscilloscope calibrator tool used in the RF Power Measurement Kit, he can determine this correction factor himself. See Appendix D for more information concerning the derivation of the correction factor. Divide the Vpp value read from the oscilloscope by the correction factor to determine the approximate true RF voltage level value. The simple formula is as follows:

$$\frac{\text{Measured Vpeak to peak}}{\text{Correction factor}} = \textbf{Approximate True RF Voltage Value}$$

E-1 Exciter RF Output Measurement Using an Oscilloscope

1. Connect the oscilloscope to the system hardware as per Illustration E1 -1. Confirm that the oscilloscope is properly configured, that the channel 1 vertical Volts/div variable control is fully CCW, and that the bandwidth limit button on the oscilloscope is not selected.



RF SIGNAL MEASUREMENTS USING A ≥ 300 MHZ OR CHARACTERIZED 100 MHZ SCOPE
ILLUSTRATION E1-1

2. Ensure that channel 1 is terminated with a 50 ohm feed-through terminator. Disconnect the cable going to J14 on the front of the RFI and connect it to the input of the 50 ohm terminator connected to channel 1 on the scope. A short (less than 5 ft.) length of coax and a BNC bullet (female BNC/female BNC) adapter can be connected to the cable removed from J14 to route the signal to the scope.

3. Prepare the system to scan using the proprietary protocol listed in Table 2-3.

4. **[Manual Prescan] [Scan TR].** Increase TG to 200.

5. If using a scope with a bandwidth of 300 MHz or greater then read the peak to peak voltage from the scope face and confirm that it is $\geq$ 0.632 Vpp (0 dBm).

6. If using a < 300 MHz scope read the peak to peak voltage from the scope face and then, *using the correction factor for the scope channel in use*, calculate the approximate true RF voltage value by using the formula below. Confirm that the approximate true RF voltage value is $\geq$ 0.632 Vpp (0 dBm).

$$\frac{\text{Measured Vpeak to peak}}{\text{Correction factor}} = \textbf{Approximate True RF Voltage Value}$$

7. Decrease TG to 0 (zero). **[Done].**

E-1 Exciter RF Output Measurement Using an Oscilloscope (Continued)

8. Reconnect the cable removed from J14 on the front of the RFI.

9. If the measured voltage meets the specification then the Exciter is working properly. The functionality of the RFI should be checked again. No need to continue this procedure.

10. Disconnect the cable going to J1 on the rear of the System Cabinet and connect to J1 a known good coax test cable 5 ft. in length. Route the other end of the test cable to the input of the 50 ohm feed-through terminator connected to channel 1 on the scope.

11. **[Manual Prescan] [Scan TR].** Increase TG to 200.

12. If using a 300 MHz bandwidth or greater scope then read the peak to peak voltage from the scope face and confirm that it is 0.893 to 1.12 Vpp (3.0 dBm to 5.0 dBm). A voltage level exceeding the 1.12 Vpp (5.0 dBm) typical upper limit is generally not a cause for concern.

13. If using a 100 MHz bandwidth scope read the peak to peak voltage from the scope face and then, *using the correction factor for the scope channel in use*, calculate the approximate true RF voltage value by using the formula in step 6 and confirm that the approximate true RF voltage value is $\geq$ 0.893 Vpp (3.0 dBm).

14. Decrease TG to 0 (zero). **[Done].**

15. Reconnect the cable removed from J1 on the rear of the System Cabinet.

16. Check for poor connections in the cabling or connectors between the System Cabinet and RF Cabinet if the voltage measured at J14 is too low but the voltage at J1 on the rear of the System Cabinet meets specification.

17. Remove connector from J109 on the bottom, front of the (U)CERD.

    a. Insert the cannon-to-BNC test cable from the “TPS RF Connector/Adapter and Cable Test Kit” into the top-most female plug inside the J109 connector on the (U)CERD, marked EXC RF OUT.

    b. Connect the BNC end of the test cable to the 50 ohm feed-through connector on scope channel 1.

18. **[Manual Prescan] [Scan TR].** Increase TG to 200.

19. If using a 300 MHz bandwidth or greater scope then read the peak to peak voltage from the scope face and confirm that it is 0.893 to 1.12 Vpp (3.0 dBm to 5.0 dBm). A voltage level exceeding the 1.12 Vpp (5.0 dBm) typical upper limit is generally not a cause for concern.

20. If using a 100 MHz bandwidth scope read the peak to peak voltage from the scope face and then, *using the correction factor for the scope channel in use*, calculate the approximate true RF voltage value by using the formula in step 6 and confirm that the approximate true RF voltage value is $\geq$ 0.893 Vpp (3.0 dBm).

E-1 Exciter RF Output Measurement Using an Oscilloscope (Continued)

21. Decrease TG to 0 (zero). **[Done].**

22. Remove the cannon-to-BNC test cable from the top-most plug of J109 on the (U)CERD and reconnect the cable that was previously removed from J109.

23. Only consider replacing the Exciter if the output is below the specification **AND** the body RF output power cannot be adjusted to meet the 16 kW specification. Otherwise, check the RF cabling, connectors, or any hardware in between.

E-2 Exciter RF Output Check Using The RF Power Measurement Kit

More accurate measurements are made by using the RF Power Measurement Kit and a 100 MHz oscilloscope. The kit will compensate for any amplitude error due to the bandwidth limitation associated with using a 100 MHz scope. The 1.5T Scope Calibrator provides a 63.86 MHz sine wave at 1.00 VPP (4dBm).

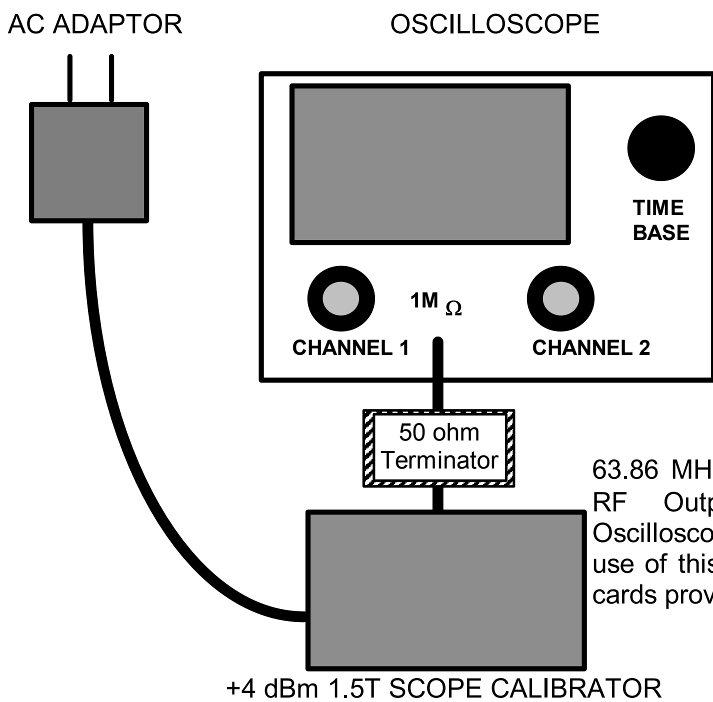
1. Verify that the Bandwidth Limit button is <u>not</u> selected on the oscilloscope.

2. Verify Channel 1 is set to 1 Mega ohm input termination.

Note

Use the 50 ohm terminator supplied with the RF Power Measurement Kit and set Channel 1 oscilloscope termination selection to 1Mega ohm.

3. Connect the 1.5T Scope Calibrator via the 50 ohm feed-through terminator to the oscilloscope input. See Illustration E2-1 or the laminated card included in the RF Power Measurement Kit that is labeled, in the upper right-hand corner, CAL.



Note
63.86 MHz oscillator with 4dBm/1.00 VP-P RF Output into 50 Ω termination. Oscilloscopes less than 300 MHz require the use of this 1.5T Scope Calibrator. All 1.5T cards provide Scope Readings in Divisions.

1.5T SCOPE CALIBRATOR CONNECTIONS
ILLUSTRATION E2-1

4. Adjust the channel 1 vertical Volts/div control until amplitude of the waveform is slightly greater than 8 divisions.

5. Adjust the channel 1 vertical Volts/div variable control to achieve exactly 8 divisions.

6. Remove 1.5T Scope Calibrator. Do not remove the 50 ohm feed-through terminator attached to the oscilloscope channel 1.

E-2 Exciter RF Output Check Using The RF Power Measurement Kit (Continued)

7. Disconnect the cable going to J14 on the front of the RFI and connect it to the input of the 50 ohm terminator connected to channel 1 on the scope. A short (less than 5 ft.) length of coax and a BNC bullet (female BNC/female BNC) adapter can be connected to the cable removed from J14 to route the signal to the scope.

8. Prepare the system to scan using the proprietary protocol listed in Table 2-3.

9. **[Manual Prescan] [Scan TR].** Increase TG to 200.

10. Confirm that the output from the J14 cable is $\geq$ 25.28 **minor divisions**.

Note

Understand that each of the 8 divisions on the scope face is composed of 5 minor divisions. Since 8 X 5= 40 total minor divisions, count the number of individual minor divisions that the RF waveform spans. Since the input signal is small, minor divisions are used in place of divisions so that a more accurate scope measurement can be made. This is discussed in detail in Appendix C.

11. Decrease TG to 0 (zero). **[Done].**

12. Reconnect the cable removed from J14 on the front of the RFI.

13. If the measured voltage meets the specification then the Exciter is working properly. The functionality of the RFI should be checked again. No need to continue this procedure.

14. Disconnect the cable going to J1 on the rear of the System Cabinet and connect to J1 a known good coax test cable 5 ft. in length. Route the other end of the test cable to the input of the 50 ohm feed-through terminator connected to channel 1 on the scope.

15. **[Manual Prescan] [Scan TR].** Increase TG to 200.

16. Read the peak to peak size of the waveform in divisions and then confirm that it is $\geq$ 35.7 **minor divisions**.

17. Decrease TG to 0 (zero). **[Done].**

18. Reconnect the cable removed from J1 on the rear of the System Cabinet.

19. Check for poor connections in the cabling or connectors between the System Cabinet and RF Cabinet if the voltage measured at J14 is too low but the voltage at J1 on the rear of the System Cabinet meets specification.

20. Remove connector from J109 on the bottom, front of the (U)CERD.

    a. Insert the cannon-to-BNC test cable from the “TPS RF Connector/Adapter and Cable Test Kit” into the top-most female plug inside the J109 connector on the (U)CERD, marked EXC RF OUT.

E-2 Exciter RF Output Check Using The RF Power Measurement Kit (Continued)

b. Connect the BNC end of the test cable to the 50 ohm feed-through connector on scope channel 1.

21. **[Manual Prescan] [Scan TR].** Increase TG to 200.

22. Read the peak-to-peak size of the waveform in divisions and then confirm that it is $\geq$ 35.7 **minor divisions**.

23. Decrease TG to 0 (zero). **[Done].**

24. Remove the cannon-to-BNC test cable from the top-most plug of J109 on the (U)CERD and reconnect the cable that was previously removed from J109.

24. Only consider replacing the Exciter if the output is below the specification **AND** the body RF output power cannot be adjusted to meet the 16 kW specification. Otherwise, check the RF cabling, connectors or any hardware in between.

APPENDIX F — DUMMY LOAD AND CABLES CALIBRATION

Description - This procedure provides directions for determining the true loss attributable to the dummy load and cables used when measuring the RF output power with a wattmeter or oscilloscope of 100 MHz bandwidth or greater. It is necessary to know and account for the actual loss these components contribute in order to accurately measure RF power. This procedure is not needed if using the RF Power Measurement Kit. The RF Power Measurement Kit has already been calibrated so that this loss is known and accounted for.

CAUTION

WHEN MEASURING HIGH FREQUENCY (IN THIS CASE, 42 OR 64 MHZ) ON ANY 100 MHZ SCOPE, THERE MAY BE AN ERROR DUE TO THE BANDWIDTH LIMITATIONS OF THE SCOPE. (IF NECESSARY, READ 'SCOPE TRIVIA' AT THE END OF THIS PROCEDURE.)

F-1 Overview

Test cables long enough to reach the cables, connectors and dummy load to be tested are connected between the Exciter RF Output (System Cabinet J1) and Receiver Body Input (System Cabinet J2). Receiver gains (R1 & R2) and transmit gain (TG) are set for near full scale reading on the power spectrum during prescan calibration. A reference scan is taken and stored in a raw file. The Attenuation Test Tool is used to calculate the baseline factor from the reference scan for the test cables (i.e., there is some loss from the test cables).

The dummy load and/or cable(s) to be tested are next inserted in series with the test cables and another scan is taken. Again, the Attenuation Test Tool is used to determine the "Magnitude Squared Attenuation Factor" (i.e., how much has the test signal been attenuated?). This attenuation factor is used in the RF power calibration process to accurately calculate the RF power level.

Note

If any problems are encountered during the following procedure, always start over at the beginning and re-do the reference scan. Then you may add, as directed in this procedure, any type of attenuation hardware you might have reason to test.

F-2 Tools and Instruments Required

See Table F2-1

TABLE F2-1
REQUIRED TOOLS AND INSTRUMENTS

Item	Description	Part Number	Qty.
1	50-ohm dummy load, 200 watt, 30 dB attenuator - Bird Model 8322 (or equivalent).	46-317724P14	1
2	RF Test Cables Kit	46-255816G1	1

F-3 Initial Setup

1. Disable the TNS.

CAUTION

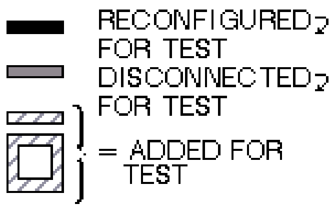
Failure to bypass the body TNS out of the circuit may result no signal being received by the system. Merely disabling the TNS by moving the Enable/Disable toggle switch down to the Disable position instead of bypassing it out of the circuit may work, however, the TNS processor can override the switch.

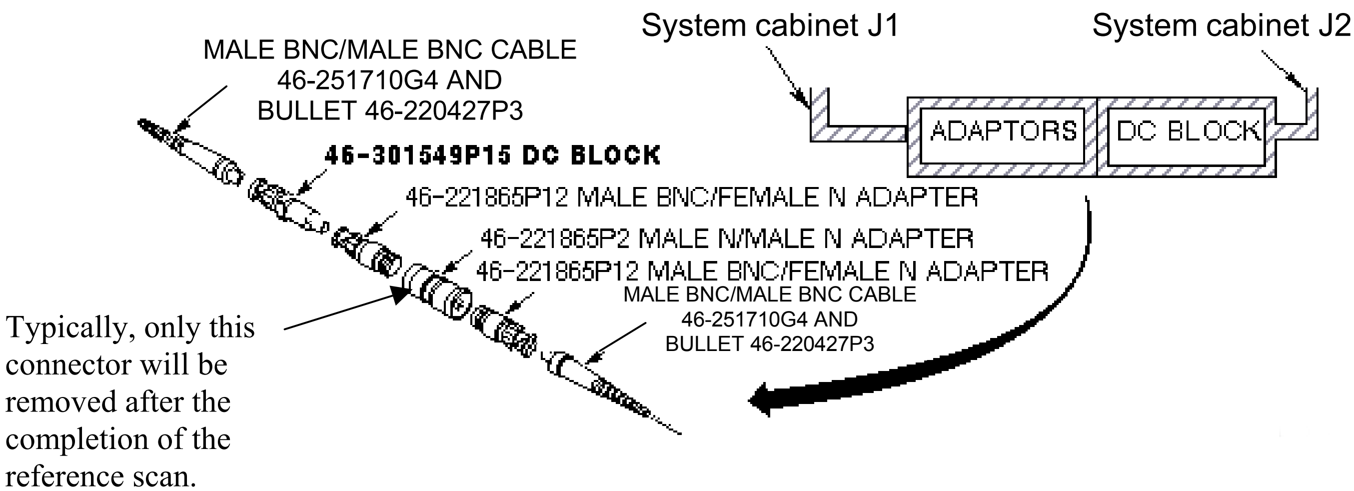
   a. Locate the body TNS on the TNS module. This is one of the two shiny metal TNS boxes affixed to the main TNS module assembly that is mounted farthest from the rear of the cabinet. Multicoil systems have an additional piggyback board affixed with 3 extra TNS boxes that mounts over the top of the main TNS module assembly.

   b. Bypass the body TNS (A24 A1 A1) from the circuit by disconnecting the small coax cables A24 A1 A1 J12 (signal output) and A24 A1 A1 J2 (signal input) from the TNS and connecting both together using a female-BNC to female-BNC adapter (also known as a bullet adapter 46-220427P3).

F-3 Initial Setup (Continued)

2. Reconfigure test hardware as shown in Illustration F3-1.





CONNECTIONS FOR INITIAL AMPLITUDE SCAN
ILLUSTRATION F3-1

Note

Adapters not shown in Illustration F3-1 can be added, if necessary, from the RF Cables Kit. Usage of the inline DC Block as shown in Illustration F3-1 is mandatory.

3. Disconnect the existing cables at the Systems cabinet I/F panel J1 (RF Out) and J2 (Receiver Body Input) and set them aside.

4. Connect the assembled test cables and adapters between J1 (RF Out) and J2 (Receiver Body Input) on the System Cabinet interface.

F-3 Initial Setup (Continued)

5. At the operator work space, prepare the system for a Dummy Load scan using the procedure, see below.

   a. Click on **[New Pt]**
      Id: **geservice**
      Name: **dummy load**
      Weight (Lb): **111**
      Set Patient Protocols to **Service**.
   b. In the Protocol field, type **o.18.1** (o=Other, 1=series number) to load the protocol.
   c. Set a landmark if necessary, then **Save Series**.
   d. With the right mouse key, select **[Research Operations]**, then select **[Display CVs]**.
      Set value of CV **calmode** to **2** (trapezoid pulse).
      (Caution here. Make sure the previous CV has been cleared before entering the next one. Look at the screen!)
      Set value of CV **p2_ramp** to **1** (1 µsec ramp time).
      Set value of CV **t2** to **50000** (50 msec tr).
      Set value of CV **pismode** to **1** (exc service).
      Set value of CV **pmode** to **1** (data collection).
      Set value of CV **daqm** to **1** (data in window).
   e. Select **[Accept]** and then select **[Research Operations]**, then select **[Download]** then select **[Manual Prescan]**.

F-4 Data Collection

1. When in **[Manual Prescan]**, set **R1** to **7**, and **R2** to **14.**

2. Adjust transmit gain (**TG**) to achieve an R1 or R2 (on IP display) of approximately 98%, without going over.

3. Select **[Done]**.

4. Select **[Scan]** (Ignore the message: `MR signal too large, reduce receiver gain.`) (Note: on the LX systems tested, the scan time starts at 13 seconds, counts down to 7 seconds, then ends. This is normal and is not cause for alarm.)

5. From the MR Tools desktop, select **[Cals/checks]** and then **[Attenuation Test]**.

6. Use **[Atten Test]** tool selection to analyze data, as shown in Table F4-1.

TABLE F4-1
DATA COLLECTION

<table>
<tr><th>Output/Prompts</th><th>Input/Comments</th></tr>
<tr><td>Last run number used was: XXXX</td><td></td></tr>
<tr><td>Please enter runfile number (XXXX): ....</td><td><Enter></td></tr>
<tr><td>Please select Locked / Unlocked file (L,U) (U):..........</td><td><Enter>
(working)</td></tr>
<tr><td>|***|
****
Average Max. magnitude Across All Views = aaaaa
Average Max. magnitude Squared = bbbbb
Average RMS Across All Views = ccccc
***</td><td></td></tr>
<tr><td>Do you want to make this run the reference(Y,N)(N):......</td><td>Y<Enter></td></tr>
<tr><td>STOP! Do not answer the next question at this point.
Continue with step 7 below.</td><td></td></tr>
</table>

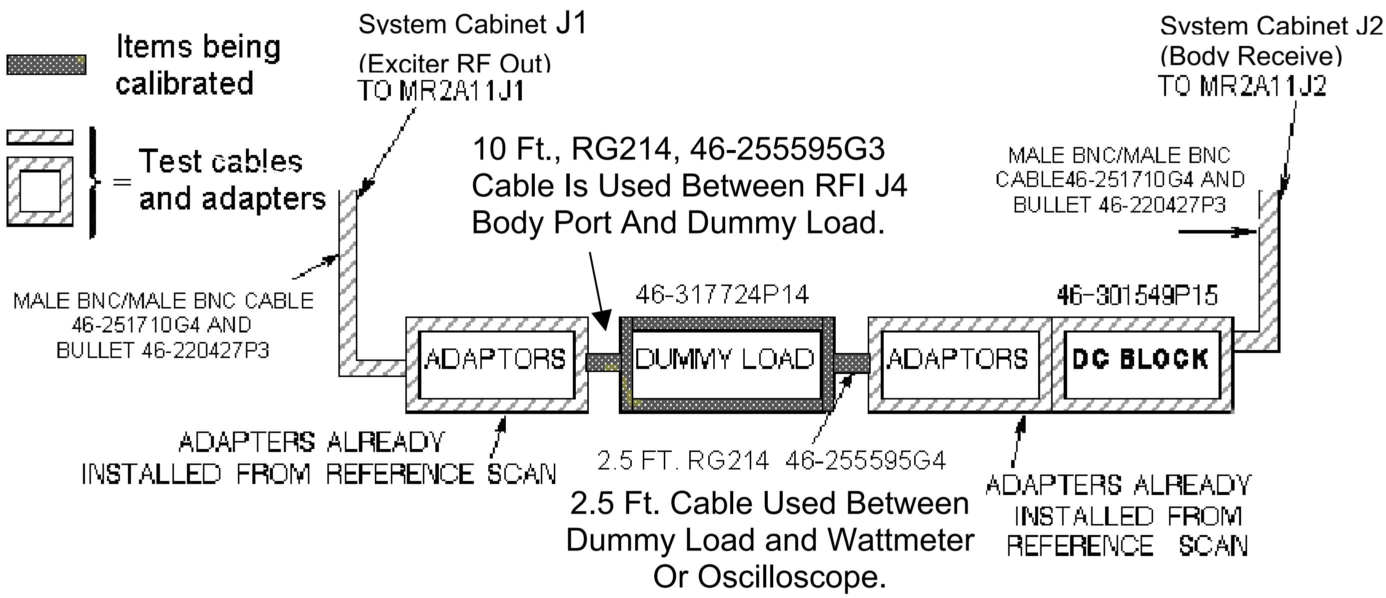
7. The next step will involve removing only the center male N to male N adapter from the test cables, setting it aside, and adding in the dummy load and cables that need to be characterized.

Note
This assumes that the item to be characterized has N female connectors at it's input and output. If it has an N connector at the input and a BNC at the output or BNC connectors at both the input and output then an additional adapter(s) will be needed in order to connect it to the test cables. Adding in one or two uncharacterized adapters should not appreciably change the baseline attenuation factor. In this case, if the wattmeter is not being used, the 2.5 ft. cable (RG214, 46-255595G4) can be eliminated from the circuit. It is not needed.

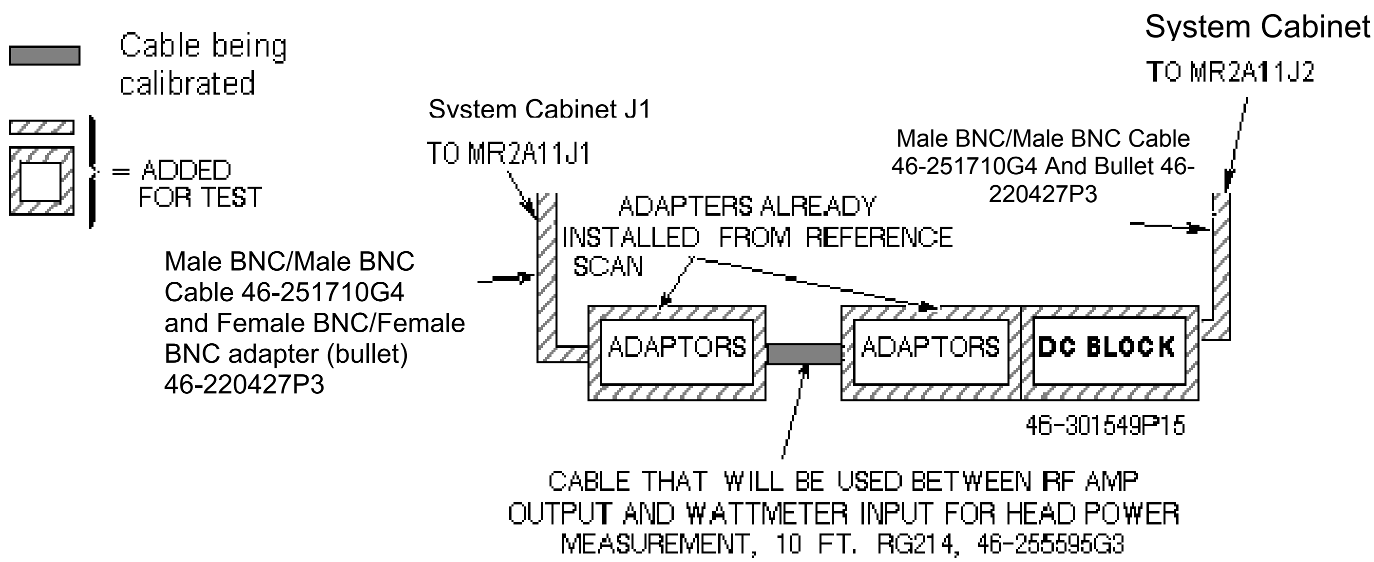
F-4 Data Collection (Continued)

8. Connect your test cables to the opposite ends of either:

- Illustration F4-1 - Dummy Load and Cables
- Illustration F4-2 - Amplifier to Wattmeter Cable.



CONNECTIONS FOR DUMMY LOAD + CABLES SCAN
ILLUSTRATION F4-1



AMP TO WATTMETER CABLE SCAN
ILLUSTRATION F4-2

Note
"Bullet" RF connector referred to in illustrations F4-1 and F4-2 is a female BNC/female BNC adapter, 46-220427P3.

F-4 Data Collection (Continued)

9. Select the scanning icon again to activate the scanning screen.

10. Select **[Scan]**.

11. When the scan is completed, re-select the tools icon again and begin the Analysis in section F-5. It will be necessary to toggle between the Scan and Toolbelt icons if multiple passes are done.

F-5 Analysis

See Table F5-1.

TABLE F5-1
ANALYSIS

Output/Prompts	Input/Comments
Do you want to compute Gain or Attenuation Ratio(G,A)[G]:	**A**<**Enter**> *IMPORTANT: Answer **after** the scan is done.*
Please do the next scan… Press <enter> key to continue, s or q to quit []:	**<Enter>**
Last run number used was: XXXX Please enter runfile number [XXXX]:.......................	<**Enter**>
Please select Locked / Unlocked file (L,U) [U]:........... |************************************| ***	<**Enter**> (Working)
Average Max. Magnitude Across All Views = aaaaa Average Max. Magnitude Squared = bbbbb Average RMS Across All Views = ccccc Magnitude Attenuation Factor = xxxxx Magnitude Squared Attenuation Factor = yyyyy	<======Record in "Value" column in Table F5-2.
RMS Attenuation Factor = zzzzz	
Do you want to make this run the reference(Y,N)(N):......	**N**<**Enter**>
***	

1. Record the "Magnitude Squared Attenuation Factor" number and record it in the appropriate Value box in Table F5-2.

TABLE F5-2
ATTENUATION FACTORS

MODE	CALIBRATED HARDWARE	PART NUMBER(S)	VALUE	NOMINAL VALUES
BODY OR HEAD	DUMMY LOAD + CABLES ATTEN FACTOR	46-255595G3 46-317724P14 46-255595G4		1000 TO 1200
HEAD	AMP TO WATTMETER ATTEN FACTOR	46-255595G3		1.0 TO 1.2

F-6 Calculation of RF power

Peak voltage should be used in the calculation in order to get an accurate result. It can be converted to power using the following formula as long as certain factors are known and accounted for. The scope correction factor MUST be known. So must the *actual* total loss attributed to anything that connects the measuring device to the source. This often includes the accumulated loss associated with the dummy load and any interconnecting cables. Table F6-1 shows the calculation of power if all the attenuating devices in the measurement circuit exhibited perfect loss; that is, the devices added no more or less loss than what they were designed to provide. Table F6-2 shows the same calculation of power but accounts for the measurement-circuit loss values in deriving the true power. Note that the loss has a significant impact on the calculated power.

WARNING!

THE SCOPE CORRECTION FACTOR MUST BE KNOWN FOR THE FORMULAE SHOWN IN TABLES F6-1 AND F6-2. IF IT IS NOT KNOWN, DO NOT USE THIS METHOD. GROSSLY INACCURATE MEASUREMENTS AND POSSIBLE SYSTEM DAMAGE WILL RESULT.

$$\frac{\left(\dfrac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{2 \text{ X } Z} \;\textbf{X}\; \text{dummy load and cables attenuation} = \text{RF Power}$$

where “X” and “**X**” in the above formula signifies multiplication

Assume Z = 50 Ω, Vpeak = 40.0, scope correction factor = 1.00 (no loss), dummy load and cables atten. = 1000, (dummy load and cables are all ideal)

$$\frac{\left(\dfrac{40.0Vp}{1.00}\right)^2}{100} \;\textbf{X}\; 1000 = 16000 \text{ Watts} = 16 \text{ kW}$$

This result assumes a ***theoretically*** ***<u>perfect</u>*** situation in which there is no loss. These situations, in common practice, rarely exist!

TABLE F6-1
RF POWER CALCULATION <u>WITH NO LOSS</u>

F-6 Calculation of RF power (Continued)

Now, consider the "real life" type situation in Table F6-2 in which the loss is considered:

$$\frac{\left(\frac{\text{Vpeak}}{\text{scope correction factor}}\right)^2}{2\text{ X Z}} \textbf{ X } \text{dummy load and cables attenuation = RF Power}$$

where “X” and “**X**” in the above formula signifies multiplication

Assume Z = 50 Ω, Vpeak = 34.61, scope correction factor = 0.88, dummy load and cables atten. = 1028

$$\frac{\left(\frac{34.61Vp}{0.88}\right)^2}{100} \textbf{ X } 1028 = 15901 \text{ Watts (very close to 16kW.)}$$

Accounting for the loss resulted in an accurate answer. 15899 Watts is as close as we can hope to get to 16000 Watts without using the RF Power Measurement Kit. Note that if none of the loss had been accounted for the error could have been **3587 Watts or 22.6%.** As a result, the observer would attempt to adjust the RF power far above the 16000 Watt limit!

TABLE F6-2
RF POWER CALCULATION WITH LOSS CONSIDERED

F-7 System Restoration

1. Reconnect original cables to the System Cabinet I/F J1 and J2.

2. Enable the TNS by removing the female-BNC to female-BNC adapter and reconnecting the RF cable A24 A1 A1 J12 (signal output) to the bottom of the body TNS module and RF cable A24 A1 A1 J2 (signal input) to the top of the body TNS module.

3. Perform one satisfactory head or body scan.

F-8 Wattmeter Trivia

Why are measurements between a oscilloscope and a Bird Wattmeter found to be different? Repeated GEMS Education Center lab groups that performed the Dummy Load and Cables attenuation process correctly, and obtained an accurate attenuation value, were able to accurately measure RF power as well as with a 400 MHZ scope! Remember, for wattmeter measurements however, the meter reading must be multiplied by the atten factor. Leaving this little factor out is the reason why the above ‘note’ was made. In body mode the wattmeter is placed after the dummy load. Its atten value must be used to calculate the correct RF value. If it is NOT used, then it should be expected there will be an error of 10% or more!

In head mode, the wattmeter is placed before the dummy load. In this case only the ‘cable’ factor is used to correct the wattmeter reading. The ‘cable’ factor is the other atten test you did for the 10 foot cable only. Typically that value is between 1.0 to 1.2. Doesn’t sound like much???? Take your calculator out for a moment. Take 2,000 watts (displayed reading) and multiply by an atten factor of 1.0 ; the results are obvious, 2,000 watts. Now take the same displayed 2,000 and multiply it by an atten factor of 1.2 = 2,400 watts! Hopefully this example will show the importance of the attenuation values. You just calculated a 20% error if you were wrong because you didn’t do atten test on the 10 foot cable!

F-9 Scope Trivia

A basic 'rule of thumb' in the scope industry is that 'to accurately measure a high frequency signal, the bandwidth of the scope should be at least 5X the frequency being measured'. That means that for a 64 MHZ signal the scope bandwidth would have to be at least 320 MHZ. Along the same line, a 42 MHZ signal would require a bandwidth of at least 210 MHZ. This is why the 400 MHZ scope was chosen several years ago for MR use. Knowing this should alert you as to what might have to be done in order to get correct scope readings from whatever scope you carry today. Who knows what bandwidth scope you'll be working with tomorrow? Will it be of sufficient bandwidth to give accurate measurements? What can you do if it is not covered by the rule of thumb? Read on!

Experimentation with multiple scopes have shown that an error is typically somewhere between 0 and 16%. That's right, some 100 MHZ scopes have no loss! Others may have as much as 15 or 16%. Additionally, it has been found that it is not necessarily the whole scope that has this loss, but the individual channels typically exhibit differences. In other words, channel one might have a 3% loss when channel two could lose 15%. There is no fixed number for all 100 MHZ scopes!

How do you get around these losses? Well, there are two ways. First you could compare a RF waveform on your scope, side by side, with a 400 MHZ scope. Use the same cables etc. and just move the cables from scope to scope. You can then calculate the loss for <u>each</u> channel of the 100 MHZ scope vs. the 400 MHZ scope. Keep the scope intensity low for best accuracy. If and when possible, determine the losses for a 1.5T and 1.0T system. That's for 64 and 42 MHZ. Point of note: there are various losses at 42 MHZ as well as with 64 MHZ. And, typically they differ! Your scope may have a 0 to 15% rolloff at 64 MHZ and 0 to 12% at 42 MHZ. All scopes differ!

Secondly, the next time you have your scope calibrated, just attach a tag asking for the % loss of signal at 42 and 64 MHZ <u>for each channel</u>. You should also state that you want this to be done at the 5 and 2 volts/div setting. This will cover both body and head measurements. At two different re-calibration vendors we use, there are no extra charges incurred for this request.

As mentioned above, the 100 MHZ scope has been shown to be as accurate as a 400 MHZ scopes if the scope rolloff factor is known. Additionally, each channel has it's own unique loss factor.

The biggest error found in GEMS Education Center labs when measuring RF power has proven to be the FE using "typical" dummy load and cables attenuation factors. Basically there are no "typical" factors! All loads and cables are somewhat different. Several may appear to be close in value, but in most cases they are different. If you use a "typical" atten factor in your calculations, then "typically" you will be in error.

Through experimentation at the GEMS Education Center, it has been determined that if you know the attenuation factor for a particular dummy load and cables, those values hold over several years. They do NOT tend to change. This means that if you could do atten test on a dummy load and cables, and keep them as a kit, you don't have to recalibrate them every time you use them. Wow! And possibly a little cheaper than the Power Measurement kit.

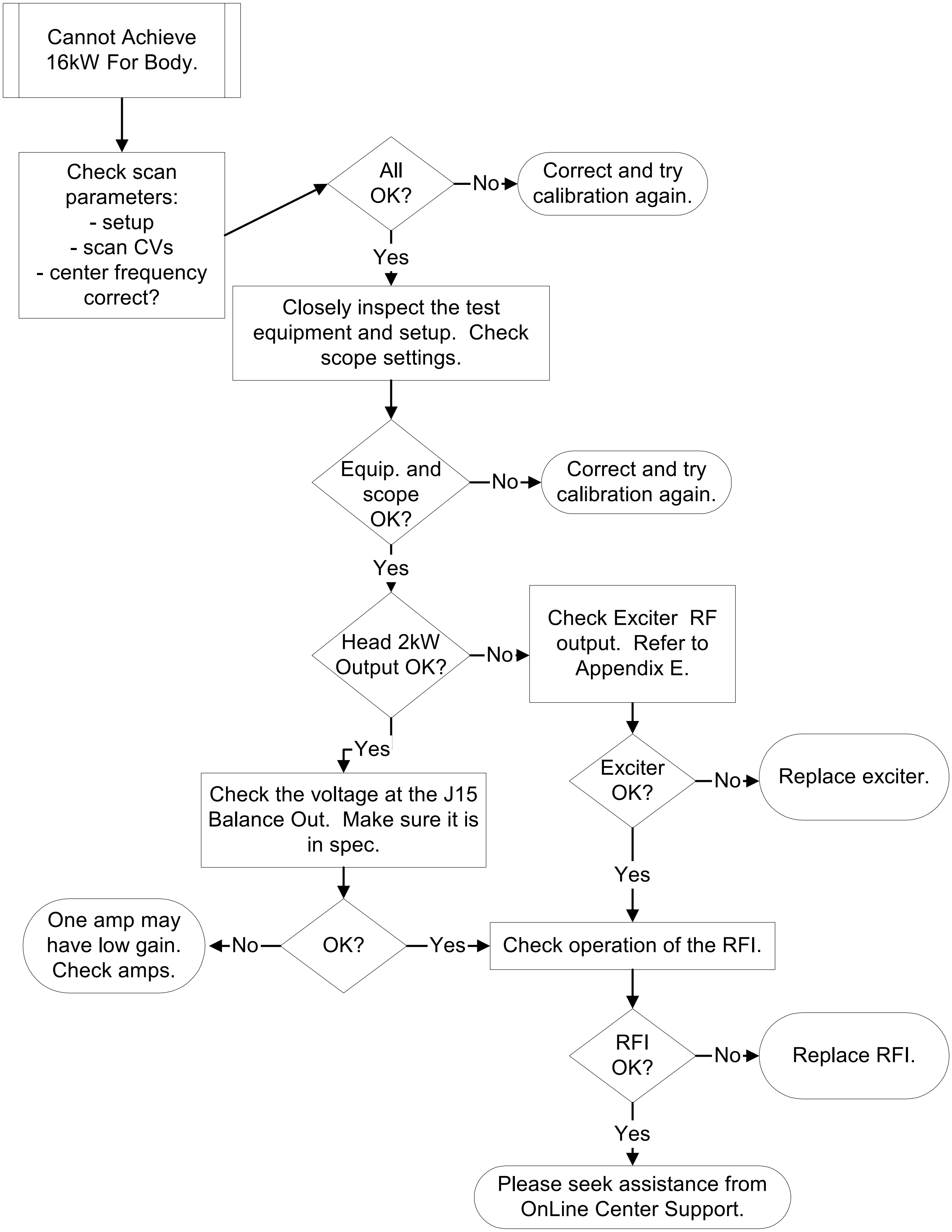
F-9 Scope Trivia (Continued)

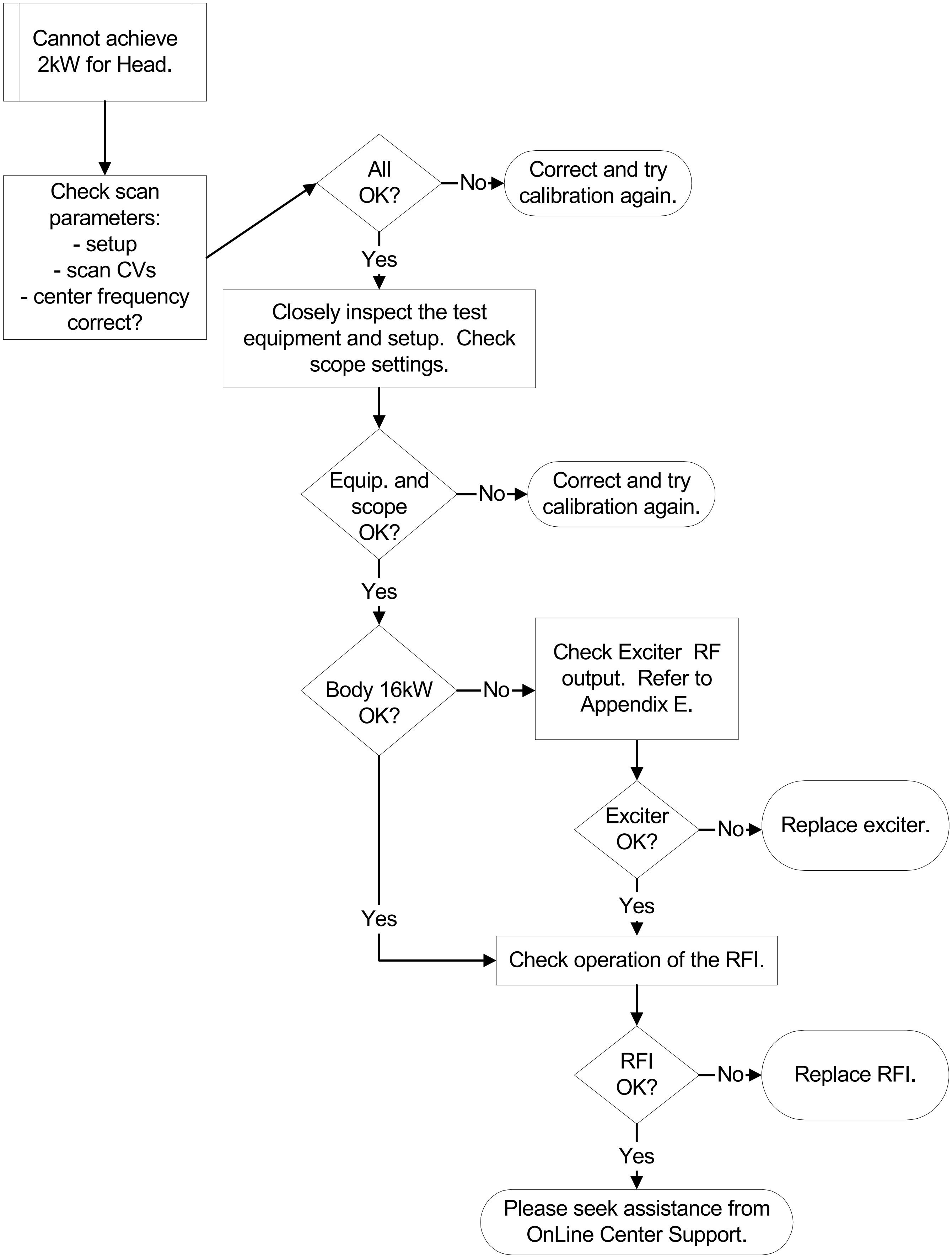
If you must, or intend to use a scope to measure body RF power, use atten test on the dummy load and cables.

Knowing the correct attenuation values and the correct scope rolloff values will provide you with measurements as accurate as a 400 MHZ scope and / or the GE Power Measurements kit.

APPENDIX G — GENERAL TROUBLESHOOTING

RF Amplifier Faults During Calibration.	→	• Reduce gain pot, try calibration adjustment again. Balance OK? See **Note** in calibration process. • Check cabling between amps and RFI. • If this is a new installation does system center frequency match magnet field strength? • Amplifiers have the same gain? Check the J15 Balance Out voltage. Make sure that both amplifiers are capable of sourcing 8.5kW.


Cannot Achieve 16kW For Body.
Check scan parameters: - setup - scan CVs - center frequency correct?
All OK?
No
Correct and try calibration again.
Yes
Closely inspect the test equipment and setup. Check scope settings.
Equip. and scope OK?
No
Correct and try calibration again.
Yes
Head 2kW Output OK?
No
Check Exciter RF output. Refer to Appendix E.
Exciter OK?
No
Replace exciter.
Yes
Check the voltage at the J15 Balance Out. Make sure it is in spec.
OK?
No
One amp may have low gain. Check amps.
Yes
Check operation of the RFI.
RFI OK?
No
Replace RFI.
Yes
Please seek assistance from OnLine Center Support.


Cannot achieve 2kW for Head.
Check scan parameters:
- setup
- scan CVs
- center frequency correct?
All OK?
No
Correct and try calibration again.
Yes
Closely inspect the test equipment and setup. Check scope settings.
Equip. and scope OK?
No
Correct and try calibration again.
Yes
Body 16kW OK?
No
Check Exciter RF output. Refer to Appendix E.
Exciter OK?
No
Replace exciter.
Yes
Yes
Check operation of the RFI.
RFI OK?
No
Replace RFI.
Yes
Please seek assistance from OnLine Center Support.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Mar 27, 1998	K. Keshena	Preliminary version.
B	June 22, 1998	K. Keshena	Second preliminary version.
0	July 28, 1998	K. Keshena	Initial Release.
1	August 13, 1998	K. Keshena	Removed scope calibration section—not applicable with 1.0T system.
2	March 15, 1999	R. Lambert (KK)	Clarified all Sections, added sections.
3	March 26, 1999	Resa Lambert	Clarified Appendix C oscilloscope use.
4	April 19, 1999	Resa Lambert	Changed RFI spec from -4 to 0 dBm, removed THS730.
5	Sept. 22, 1999	Resa Lambert	Changed TR to “55” because this value keeps changing with software (Appendix), removed THS720.
7	October 12, 2000	Don Thome'	Removed references to 1.0T amplifier, added head and body check procedure to section 2, added new flowchart, added measurement procedure and formula for oscilloscope to sections 2 and 3, added formulas to Appendix C and created Appendices D, E, F and G. Added reference to SRF Cabinet.
8	January 10, 2001	Don Thome’	Added check to t-shooting flowcharts. Changed format. Removed references to SRF.
9	March 13, 2001	Don Thome’	Corrected specification for J15 voltage. Corrected reference to table in Appendix E.
10	July 6, 2001	Don Thome’	Added reference to Power Calculator spreadsheet throughout document. Corrected various errors.
11	August 30, 2001	Don Thome’	Added comments indicating that it is no longer necessary to change ia_rf1 and ia_rf2 with 9.X software or its equivalent.
12	October 24, 2001	Don Thome’	Added comments indicating that it is no longer necessary to change ia_rf1 and ia_rf2 with 9.X, CNV4 software or its equivalent.
13	November 2, 2001	Don Thome’	Added references to RRF, UTNS, and UCERD. Added clarifications and corrections to Notes.
14	December 4, 2001	Don Thome’	Removed references to RRF and UTNS.
15	March 11, 2002	Don Thome’	Added Caution to check center frequency at start. Modified how TG was increased to improve process.
16	November 11, 2002	Don Thome’	Added “Prepare to Scan” item E to Tables 2-4 and 3-2.